

FFT 설계 & 검증 Project

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AI 시스템 반도체 설계(2기)

1조 고종완 권희식 장찬원 서윤철

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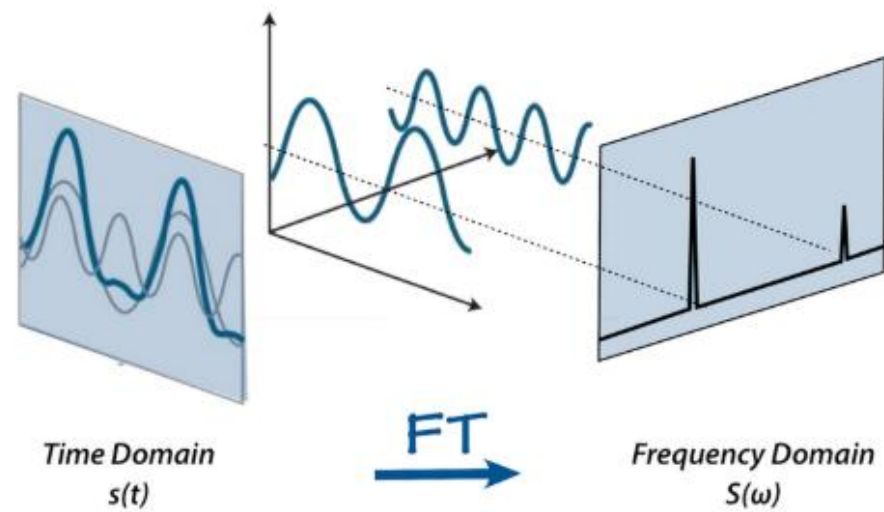
07

Trouble Shooting

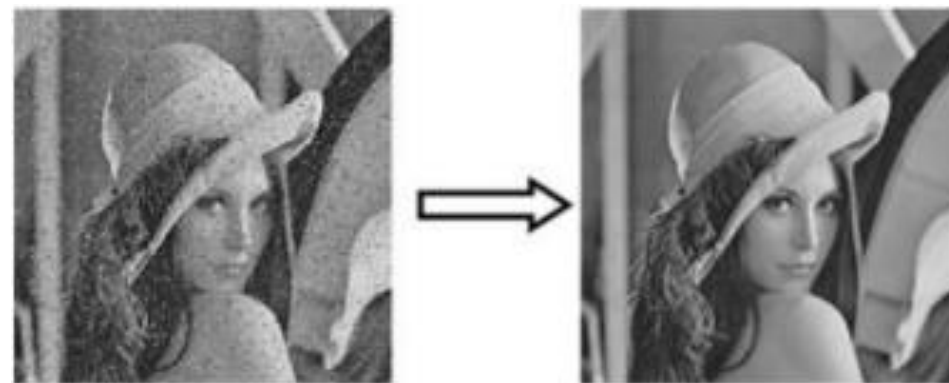
08

고찰

FFT (Fast Fourier Transform)



고속 푸리에 변환(FFT, Fast Fourier Transform)은 이산 푸리에 변환(DFT, Discrete Fourier Transform)을 효율적으로 계산하기 위한 알고리즘



영상처리



OFDM



PART 1. Floating Point Model & Fixed Point Model

FFT (Fast Fourier Transform)

Floating-point FFT

•목적: 알고리즘 검증, 기능 구현

•특징:

- 정밀도(Precision)가 매우 높음.
- 오버플로(Overflow), 언더플로(Underflow) 문제 적음.
- 소프트웨어 기반 시뮬레이션(MATLAB, Python)에서 정확한 기준값 제공.

•용도:

- 알고리즘의 동작을 확인.
- Fixed-Point 설계 전에 레퍼런스 모델로 사용.

Fixed-point FFT

•목적: 하드웨어 구현

•특징:

- 정수 + 소수 부분을 비트 폭 제한으로 표현.
- 오버플로, 스케일링(Shift) 관리 필요.
- FPGA, ASIC 설계에서 자원(LUT, DSP) 최적화 가능.

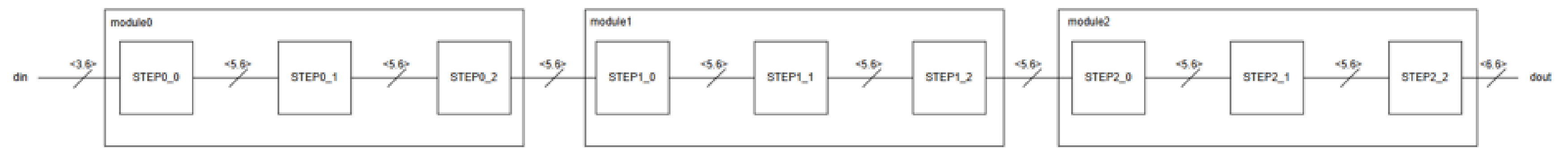
•용도:

- RTL 설계, FPGA/ASIC 구현.
- 연산 복잡도를 줄여 고속 연산 가능.

FFT (Fast Fourier Transform)

MATLAB

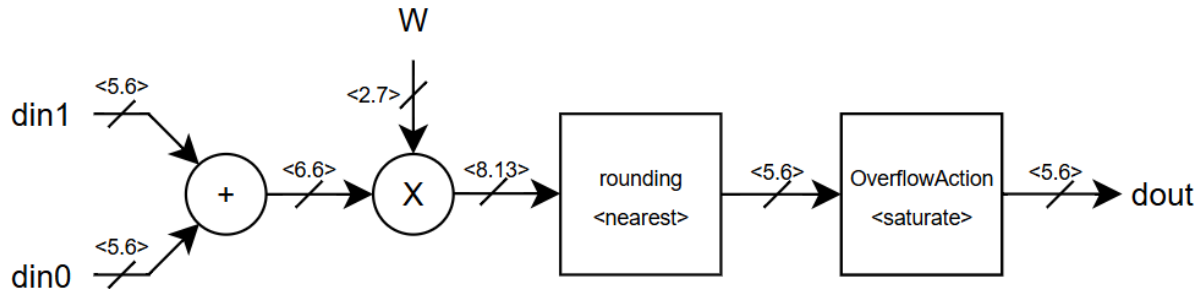
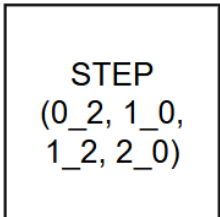
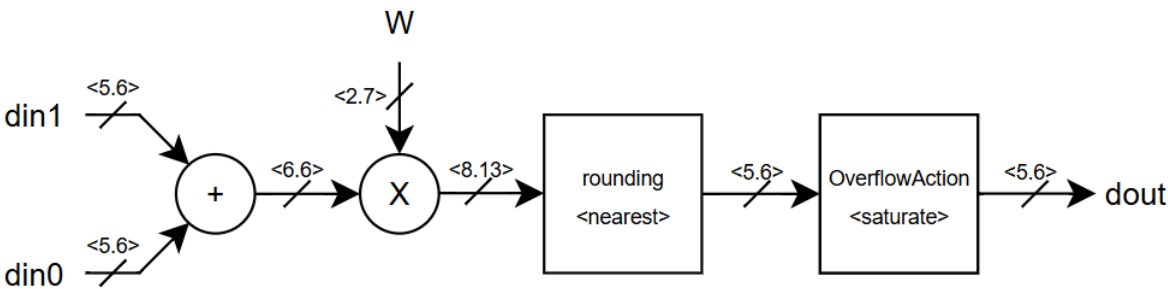
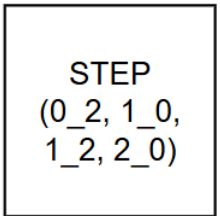
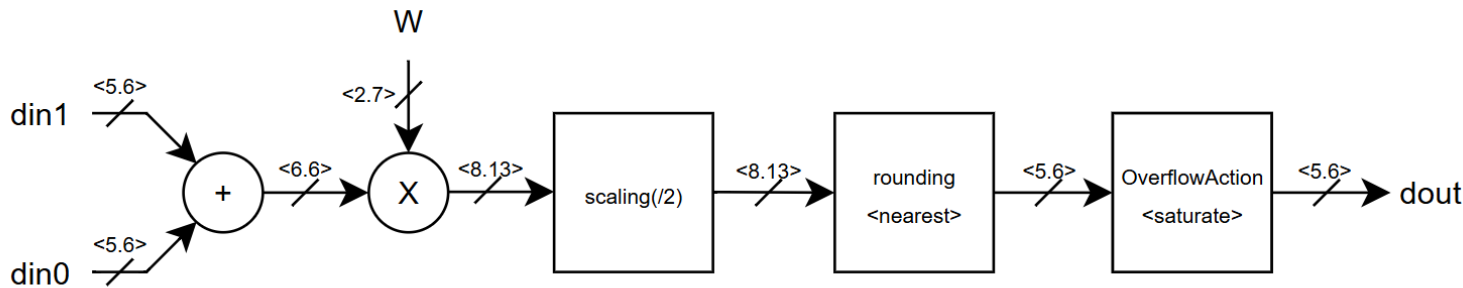
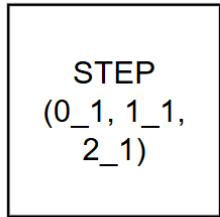
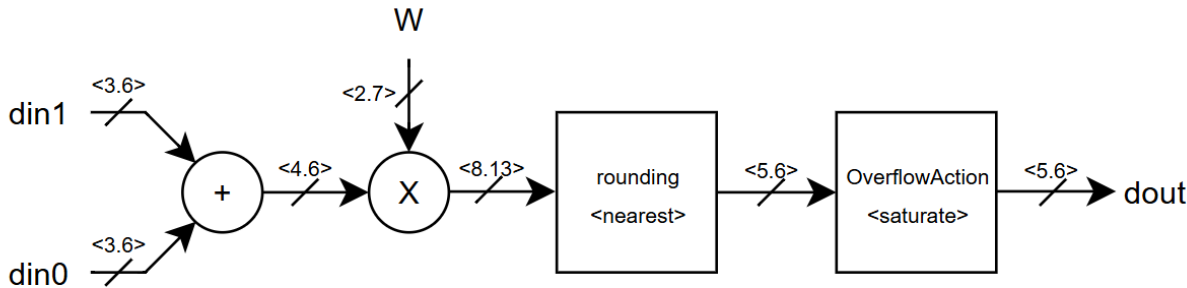
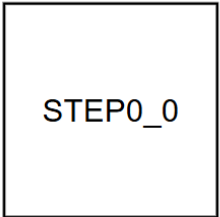
FFT BLOCK DIAGRAM



FFT (Fast Fourier Transform)

MATLAB

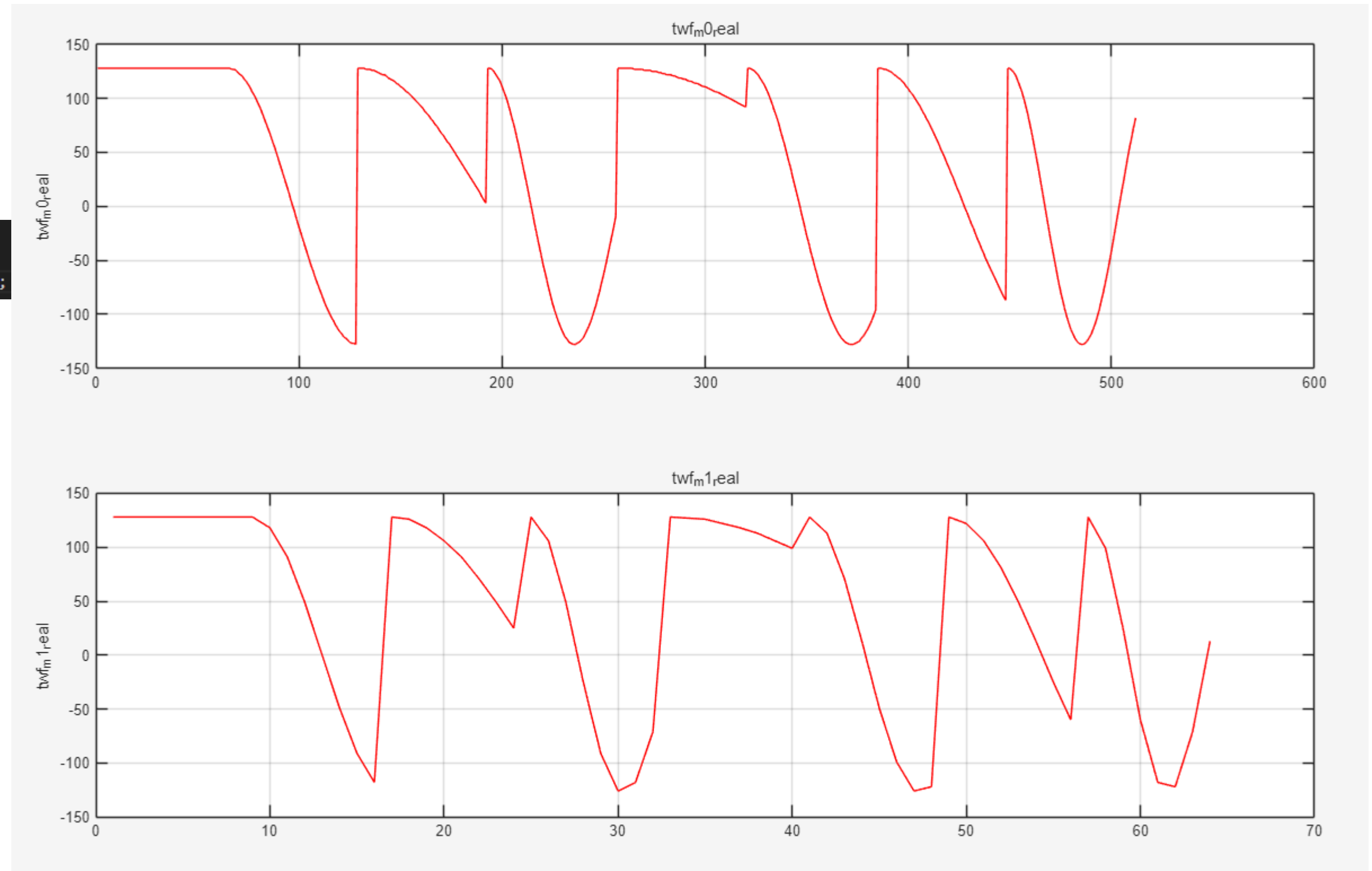
FFT BLOCK DIAGRAM



FFT (Fast Fourier Transform)

MATLAB

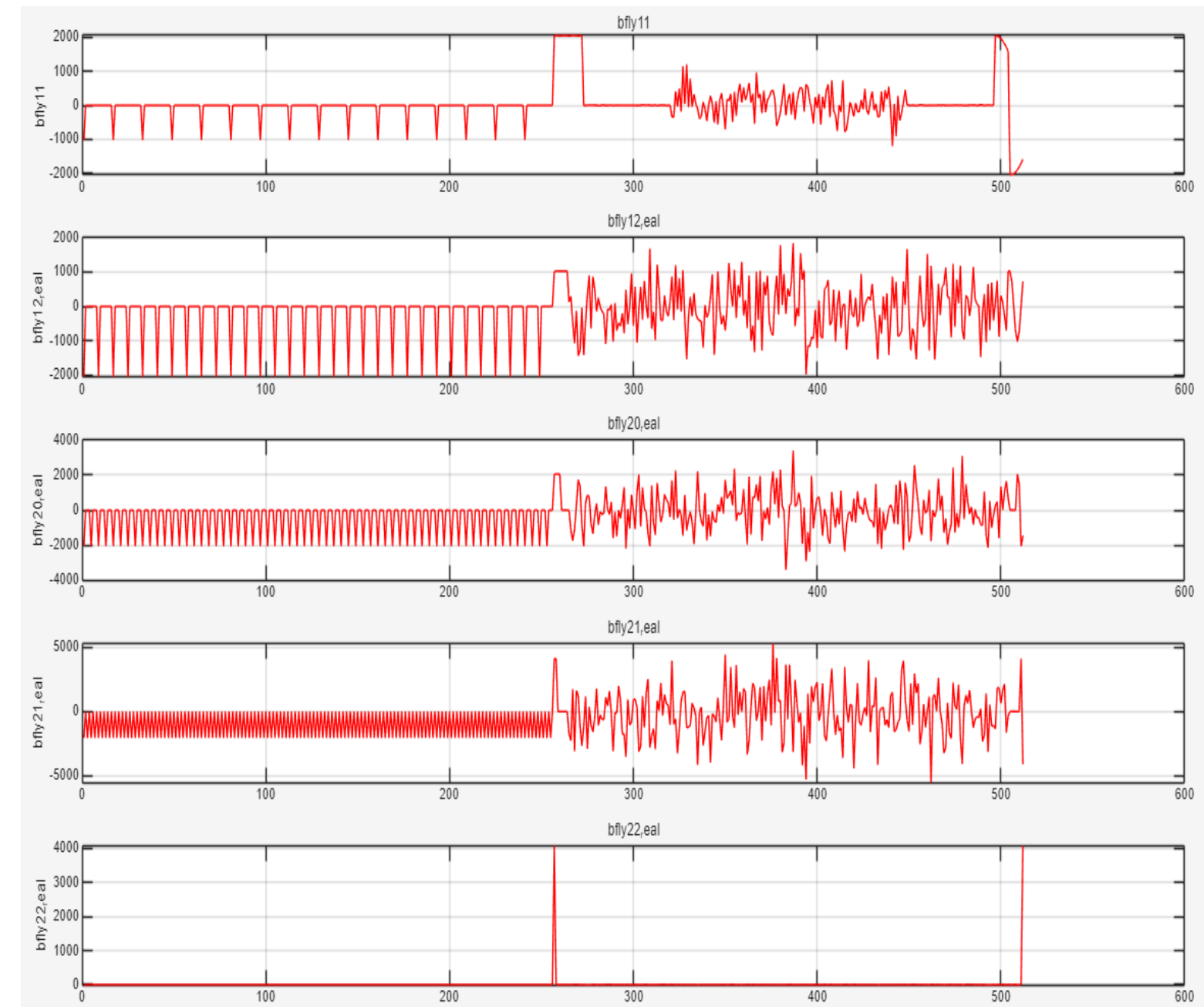
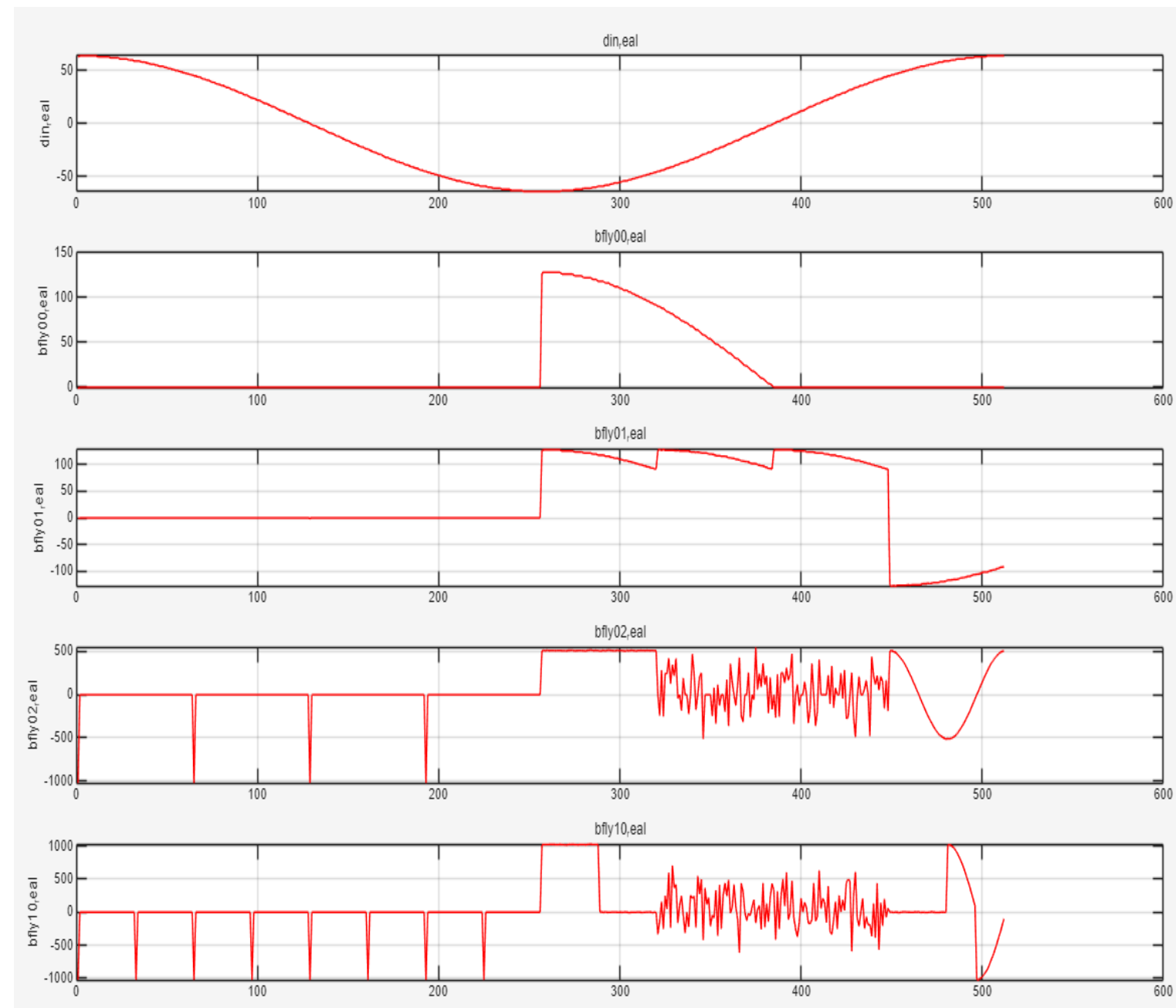
```
fac8_0 = [1, 1, 1, -j];  
fac8_1 = [256, 256, 256, -j*256, 256, 181-j*181, 256, -181-j*181];
```



FFT (Fast Fourier Transform)

MATLAB

Fixed-point FFT



SQNR(Signal-to-Quantization Noise Ratio)

양자화 오류의 정도를 나타내는 지표

SQNR이 높을수록 양자화 잡음이 적고, 원본 신호와 양자화된 신호 유사도 증가

SQNR과 양자화 잡음

양자화 잡음은 아날로그 신호를
디지털로 변환할 때 발생하는 오차

신호의 세부 사항이 소실되거나
오차가 발생하는 과정에서 발생

SQNR과 CBFP

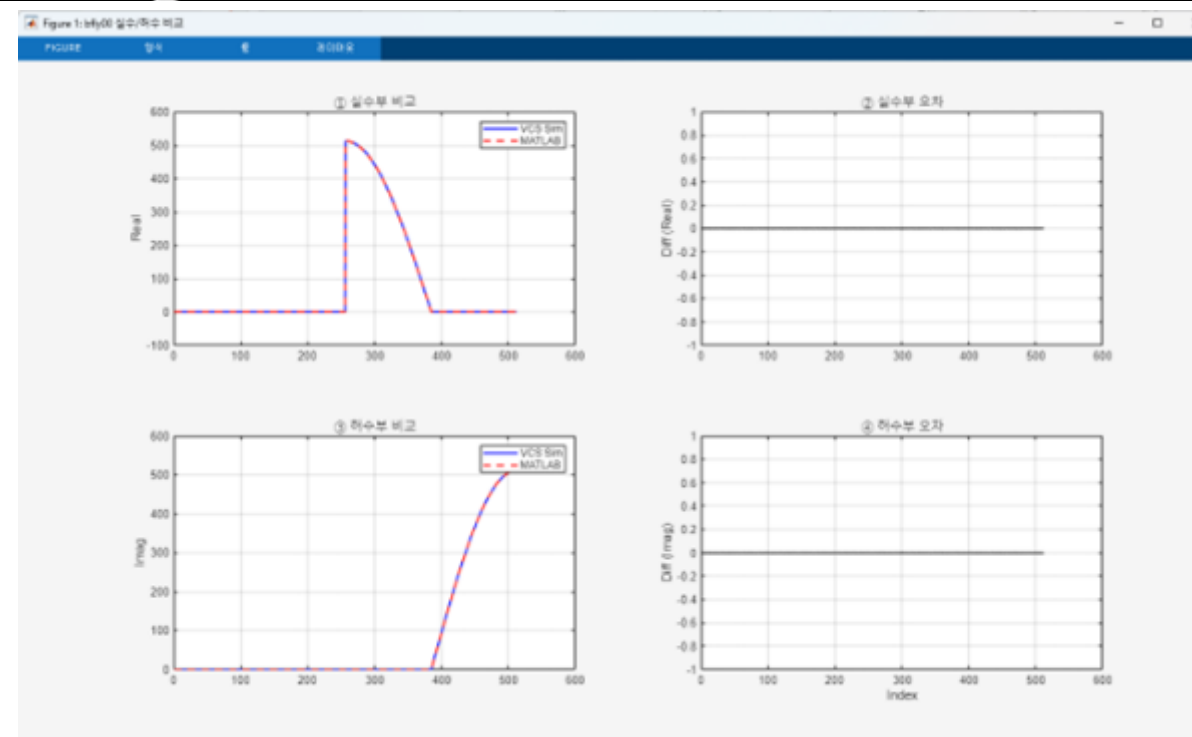
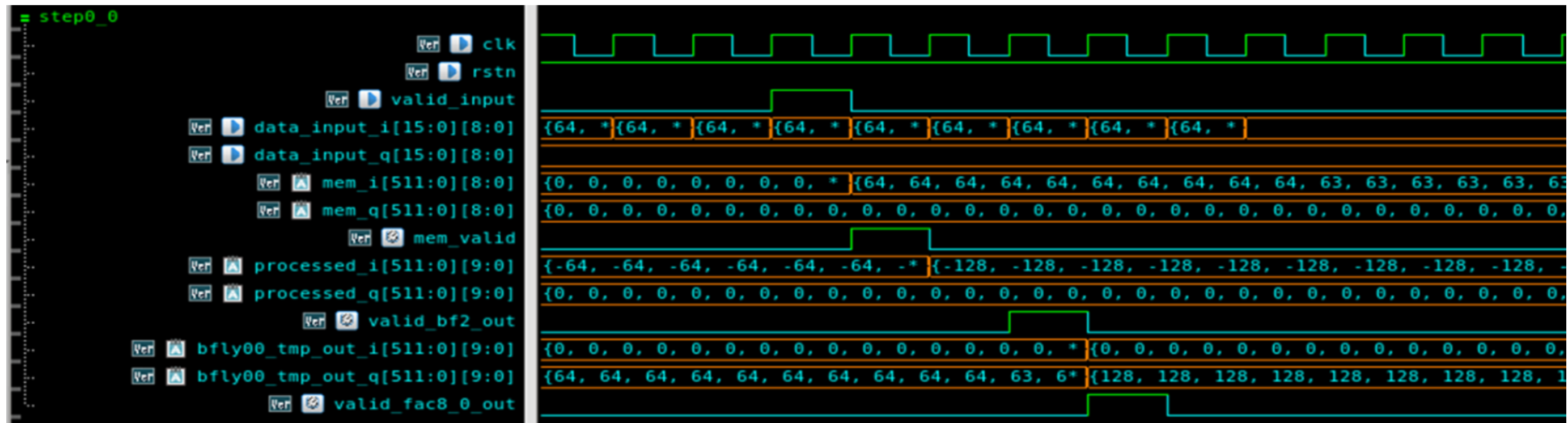
CBFP는 고정소수점 연산에서 발생하는
양자화 오류를 줄이기 위해 사용

CBFP는 동적 비트폭 조절을 통해
양자화 오류를 줄이고, SQNR을 향상

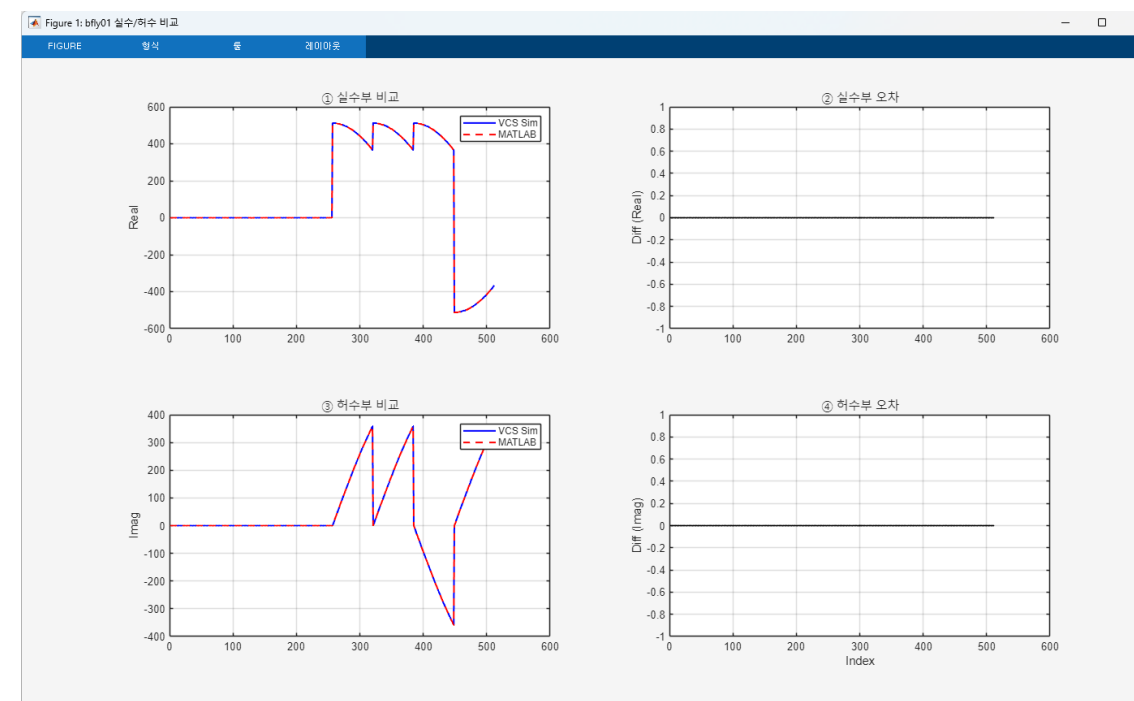
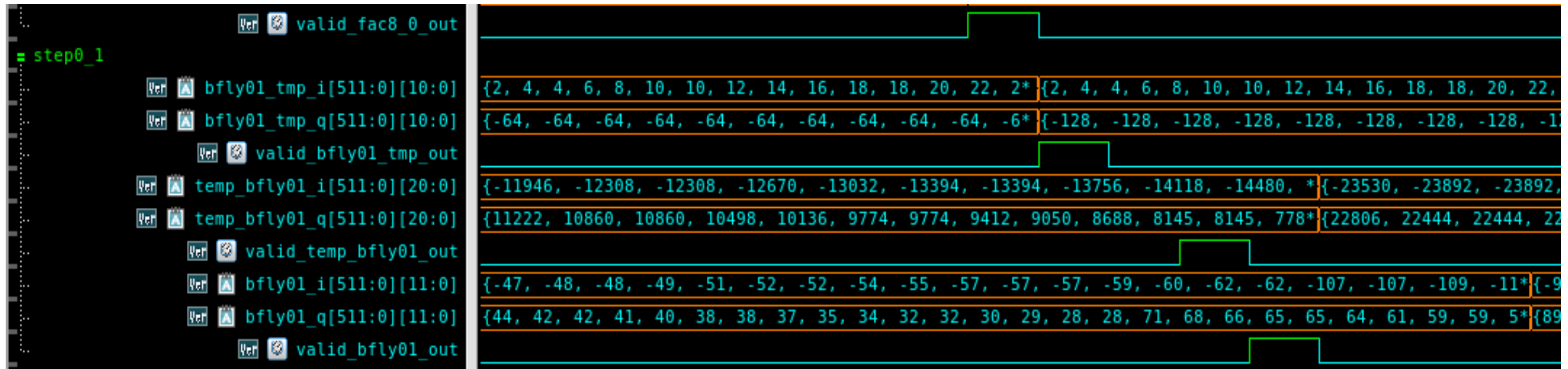
PART 3. Module Simulation & Verification

Cosine Data

Module0 – step0_0



Module0 – step0_1



Module0 – step0_2

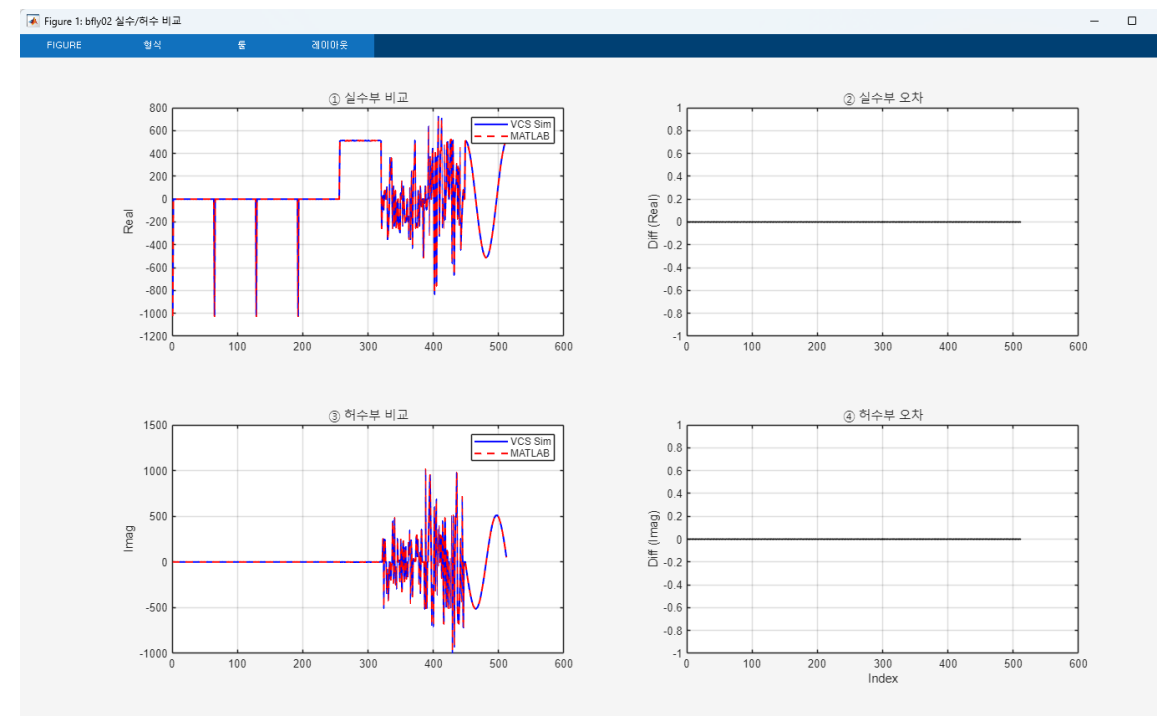
```

Ver  valid_bfly01_out

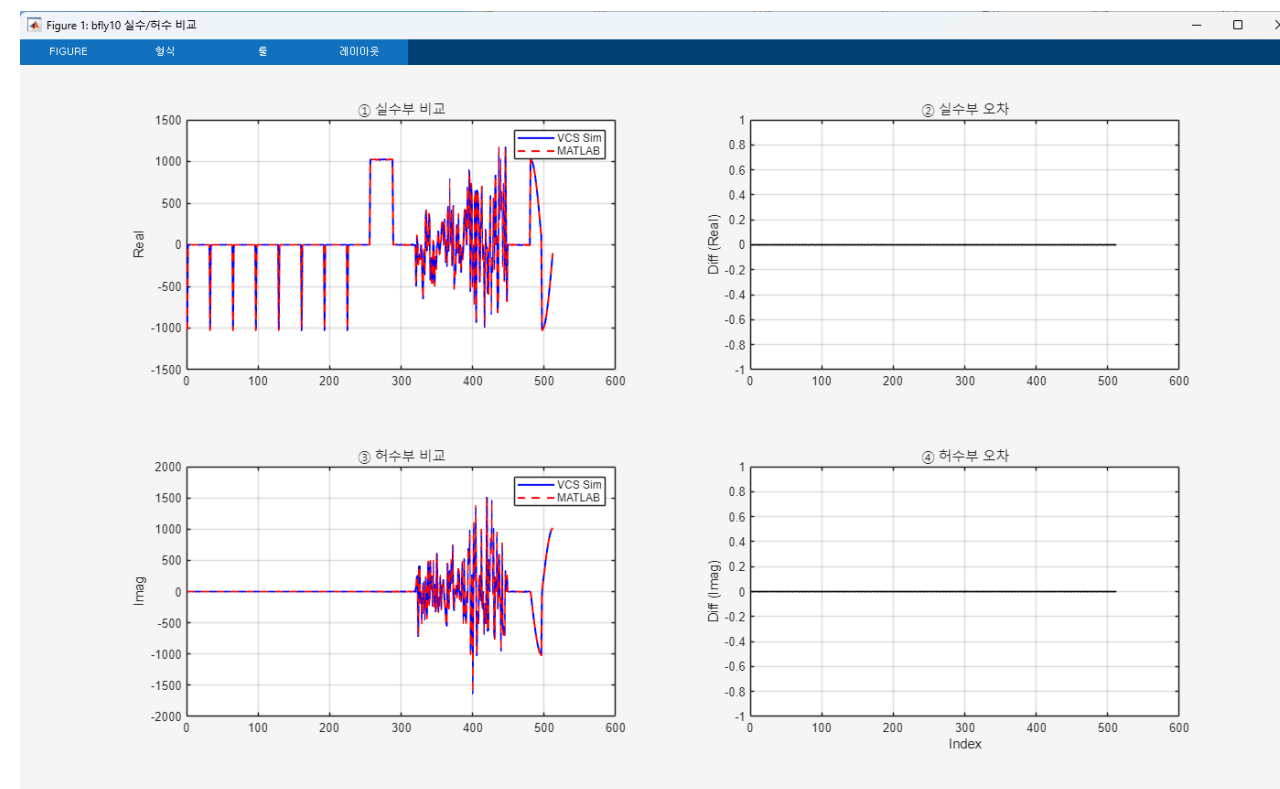
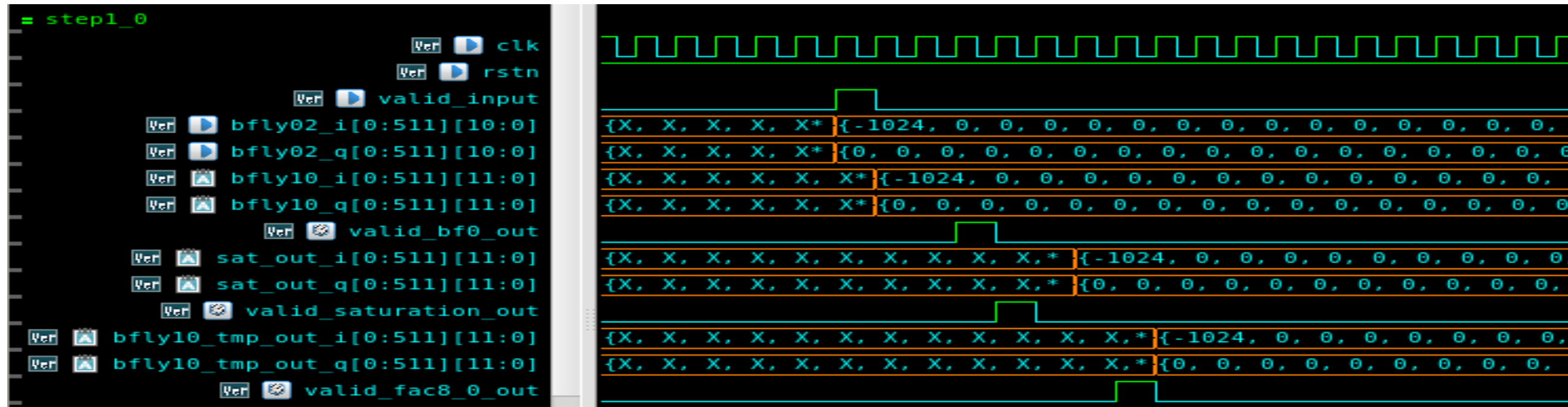
= step0_2

Ver  bfly02_tmp_i[511:0][12:0]
Ver  bfly02_tmp_q[511:0][12:0]
Ver  valid_bfly02_tmp_out
Ver  sat_out_i[511:0][12:0]
Ver  sat_out_q[511:0][12:0]
Ver  valid_saturation_out
Ver  pre_bfly02_i[511:0][23:0]
Ver  pre_bfly02_q[511:0][23:0]
Ver  valid_mul_twf_m0_out
Ver  pre_bfly02_real_buf[0:511][22:0]
Ver  pre_bfly02_imag_buf[0:511][22:0]
Ver  bfly02_i[511:0][10:0]
Ver  bfly02_q[511:0][10:0]
Ver  valid_output

```

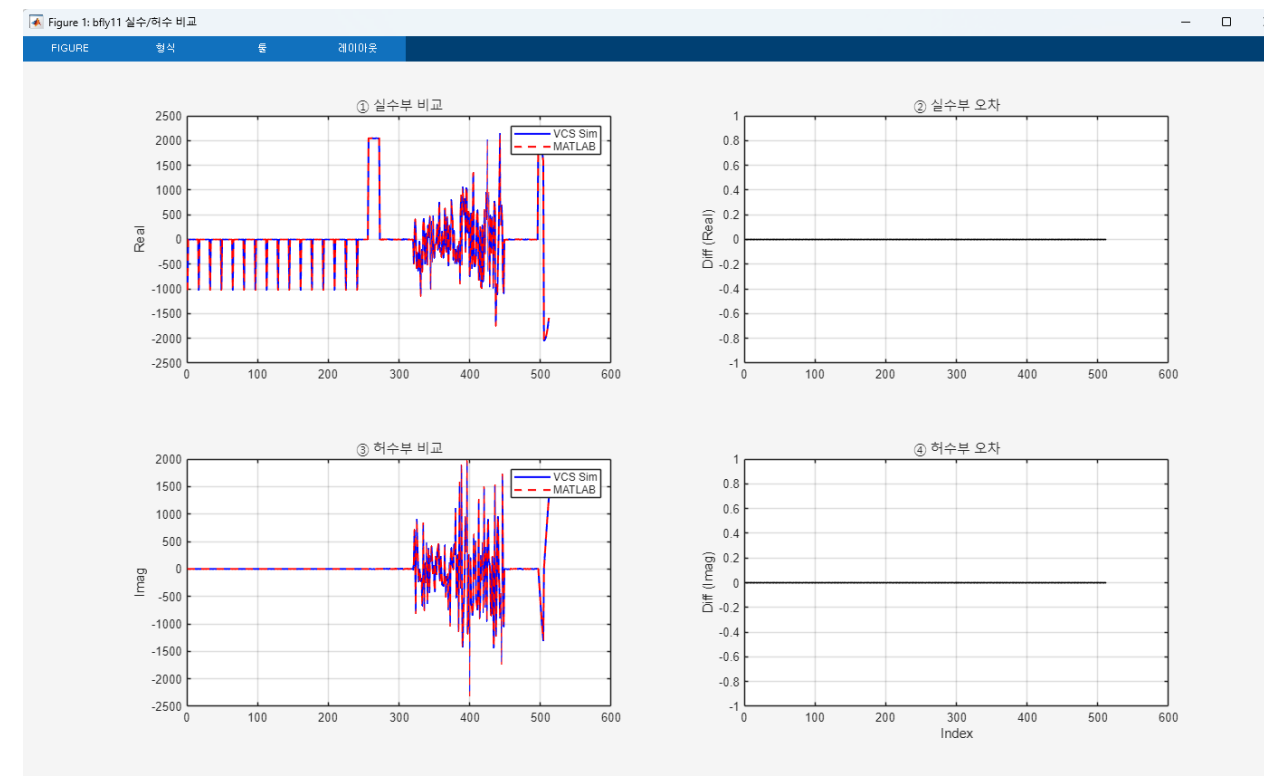
[illegible]

Module1 – step1_0



Module1 – step1_1

		valid_fac8_0_out	
	=	step1_1	
		bfly11_tmp_i[0:511][12:0]	{X, X, X, X, X, X, X, X,*}{-1024, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
		bfly11_tmp_q[0:511][12:0]	{X, X, X, X, X, X, X, X,*}{0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
		valid_bfly11_tmp	
		temp_bfly11_i[0:511][22:0]	{X, X, X, X, X, X, X, X, X, X, X, X,*}{-262144, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
		temp_bfly11_q[0:511][22:0]	{X, X, X, X, X, X, X, X, X, X, X, X,*}{0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
		valid_temp_bfly11_out	
		bfly11_i[0:511][13:0]	{X, X, X, X, X, X, X, X, X, X, X, X, X,*}{-1024, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
		bfly11_q[0:511][13:0]	{X, X, X, X, X, X, X, X, X, X, X, X, X,*}{0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
		valid_bfly11_out	

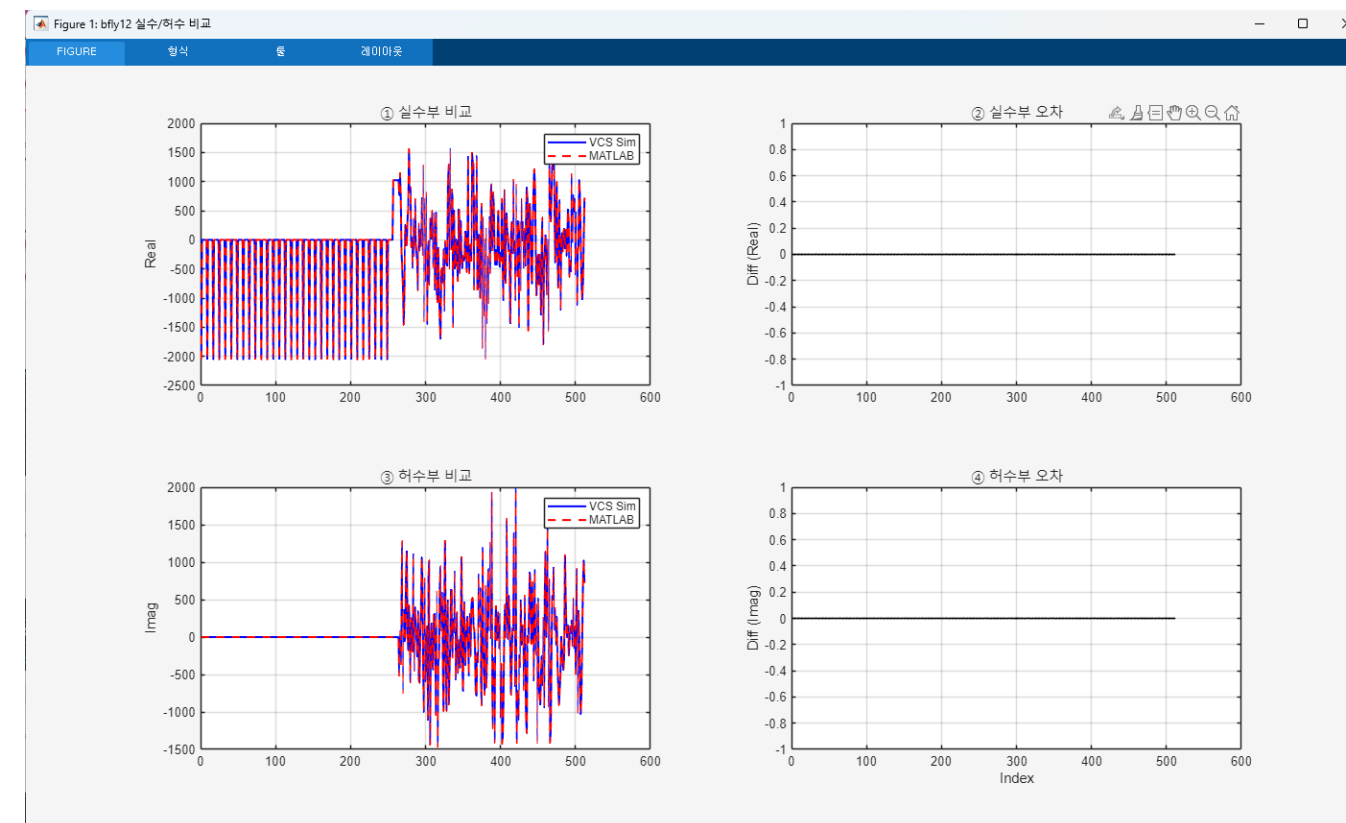


Module1 – step1_2

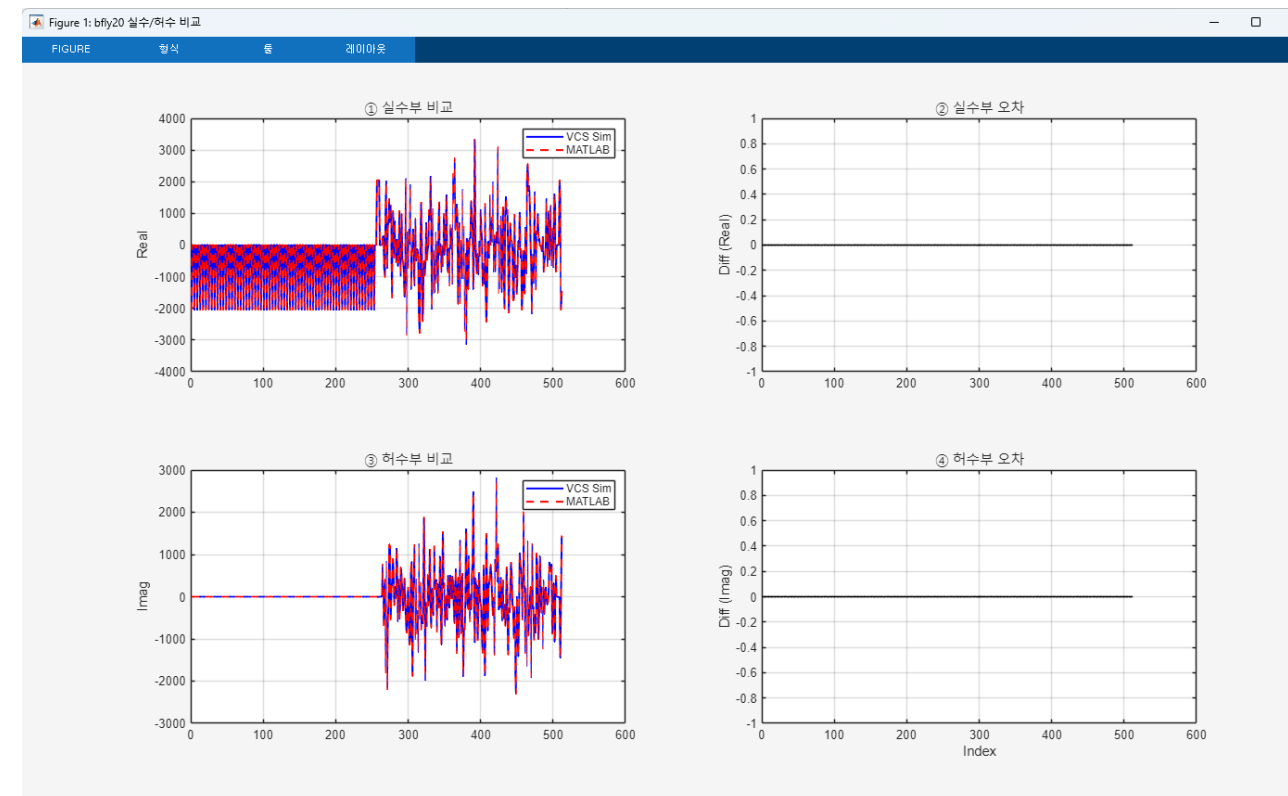
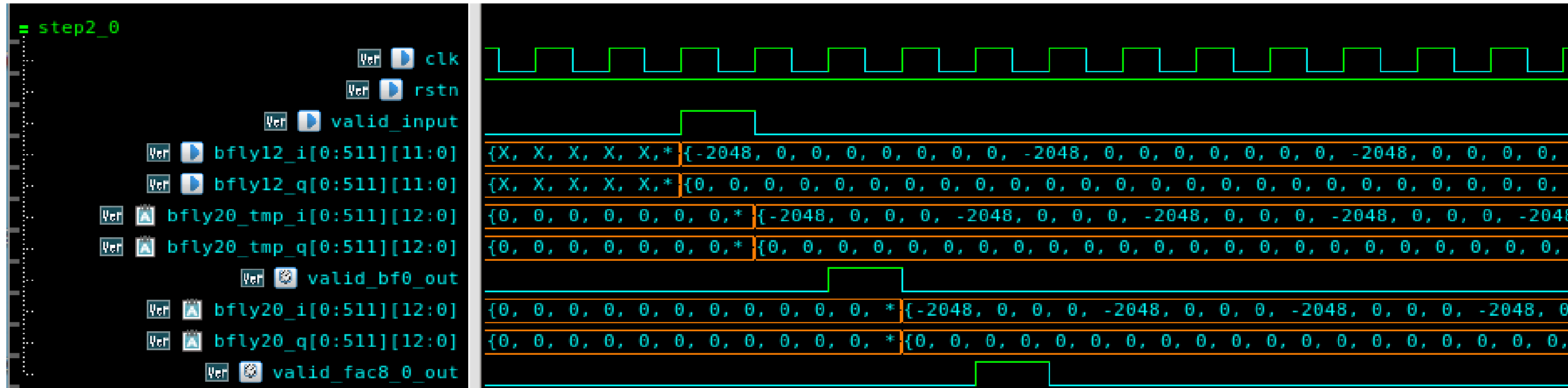
```

Ver  valid_bfly11_out
= step1_2
Ver  bfly12_tmp_i[0:511][14:0]
Ver  bfly12_tmp_q[0:511][14:0]
Ver  valid_bfly12_tmp_out
Ver  pre_bfly12_i[0:511][24:0]
Ver  pre_bfly12_q[0:511][24:0]
Ver  valid_mul_twf_m1_out
pre_bfly12_real_buf[0:511][24:0]
pre_bfly12_imag_buf[0:511][24:0]
Ver  bfly12_i[0:511][11:0]
Ver  bfly12_q[0:511][11:0]
Ver  valid_output

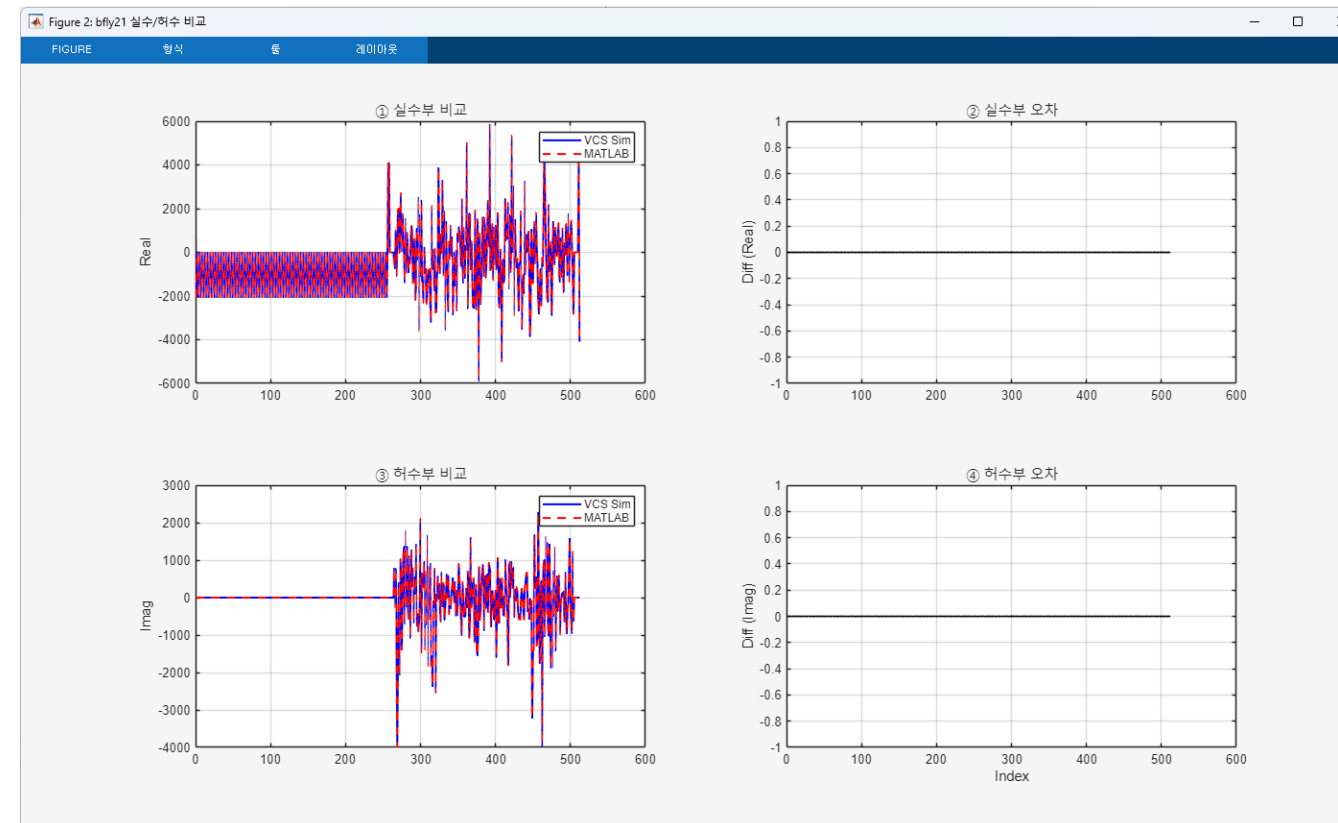
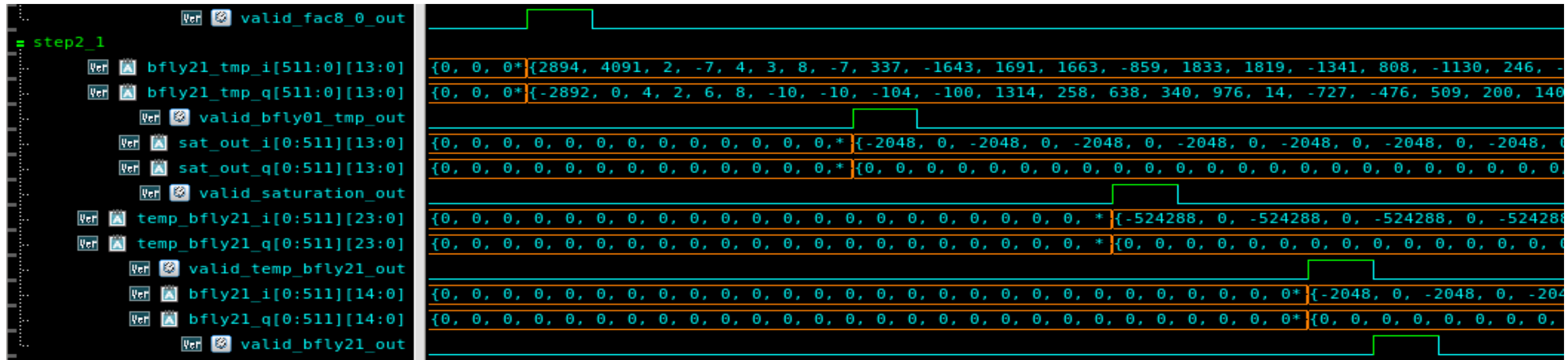
```



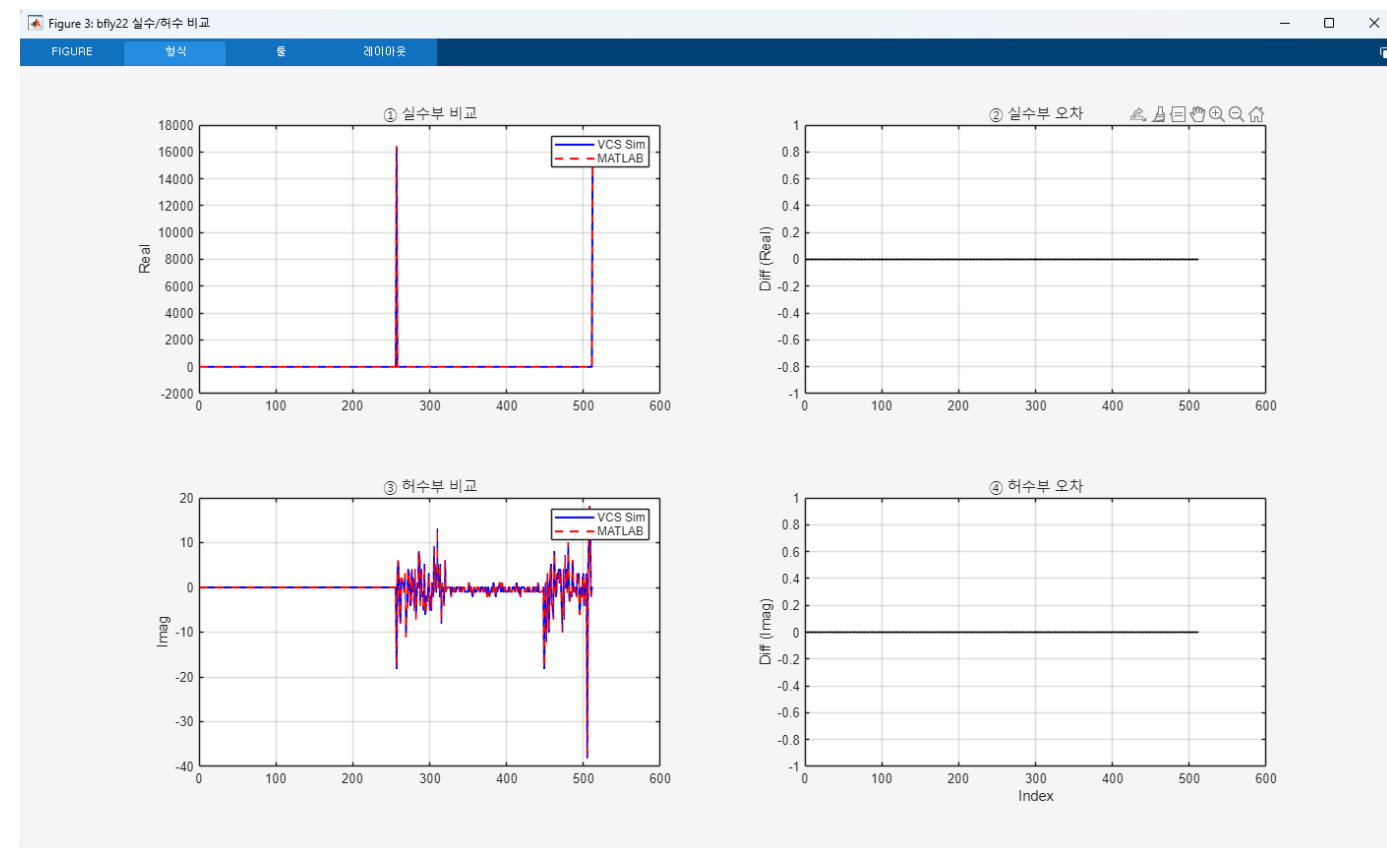
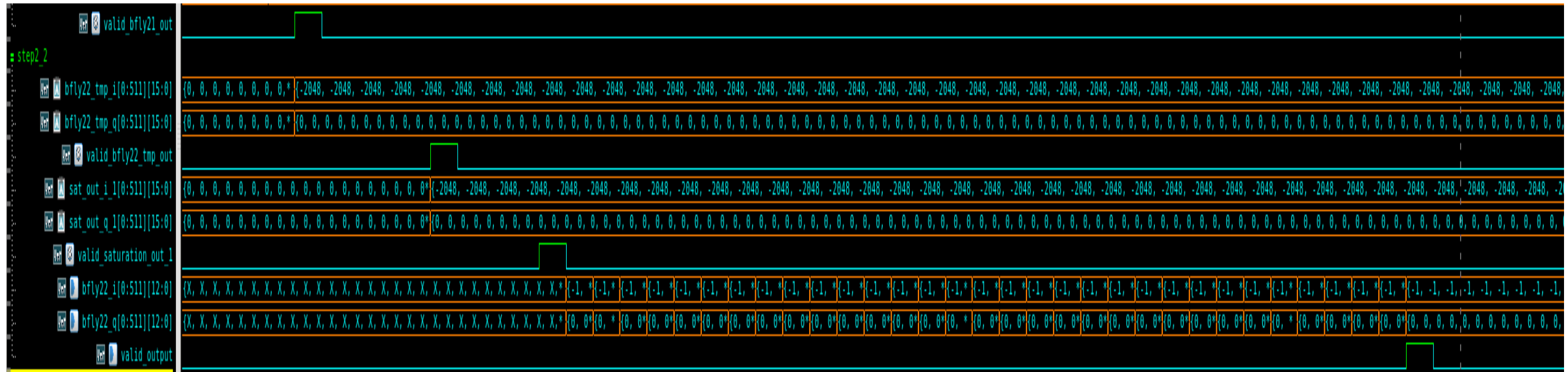
Module2 – step2_0



Module2 – step2_1



Module2 – step2_2

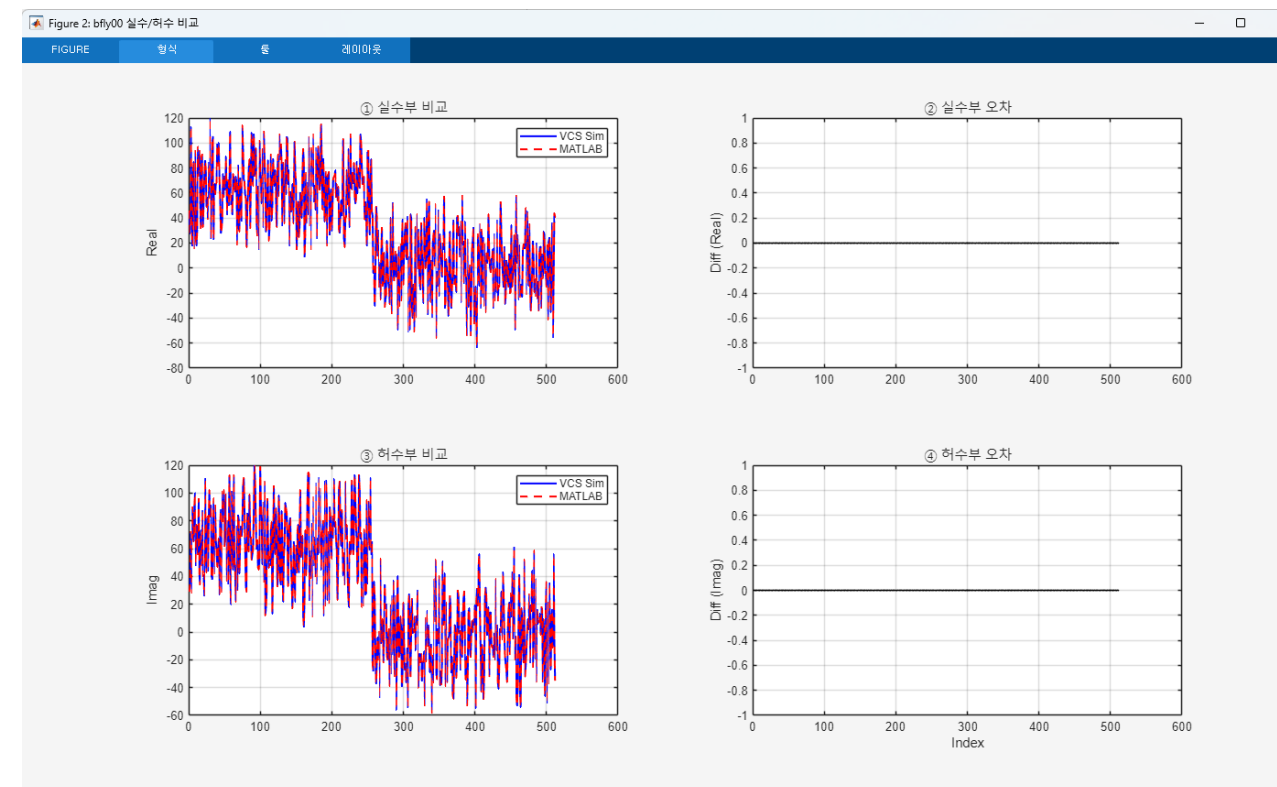
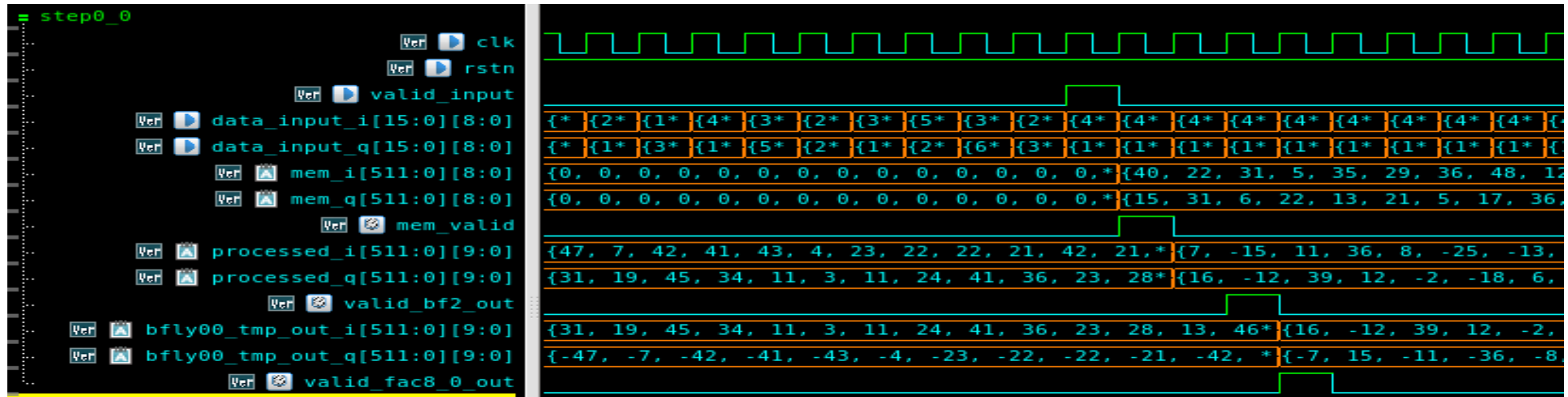




PART 4. Module Simulation & Verification

Random Data

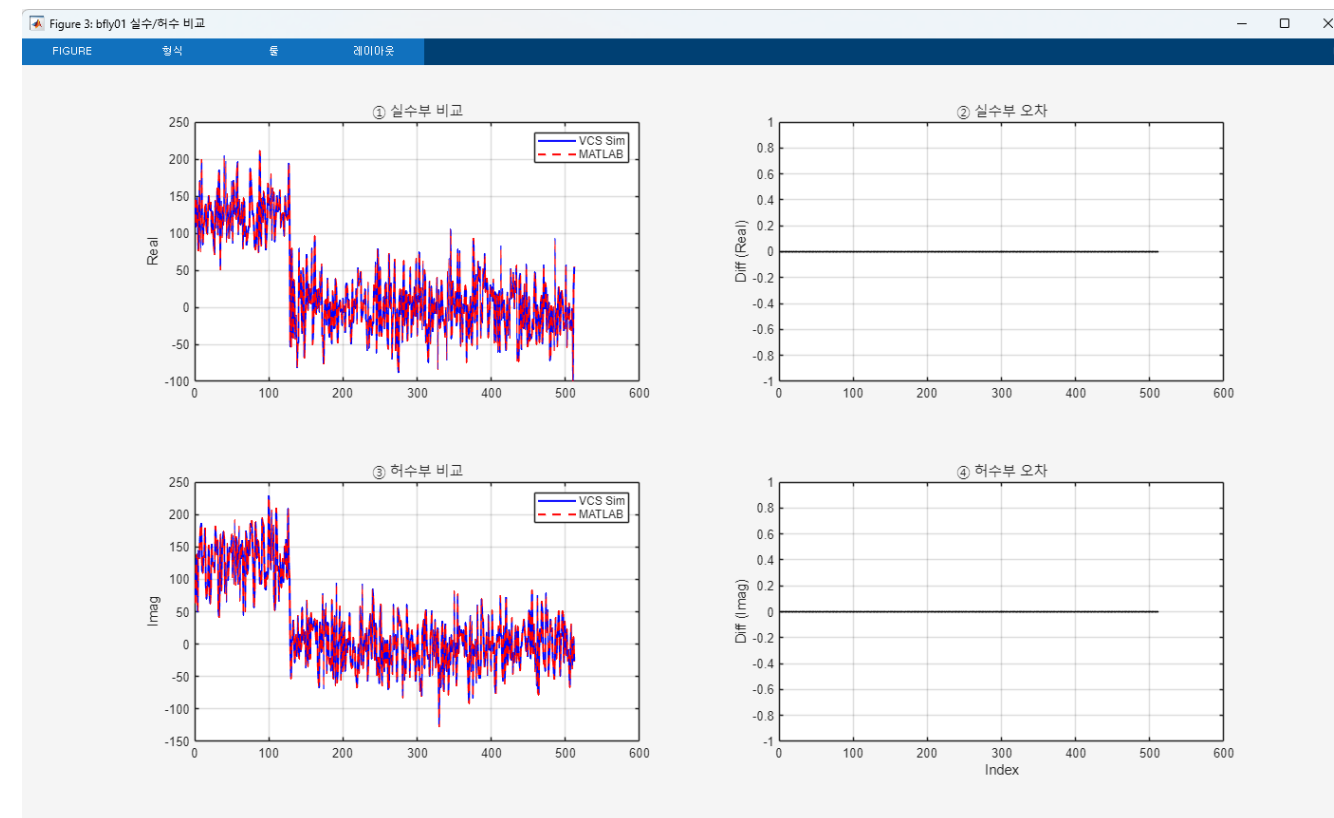
Module0 – step0_0



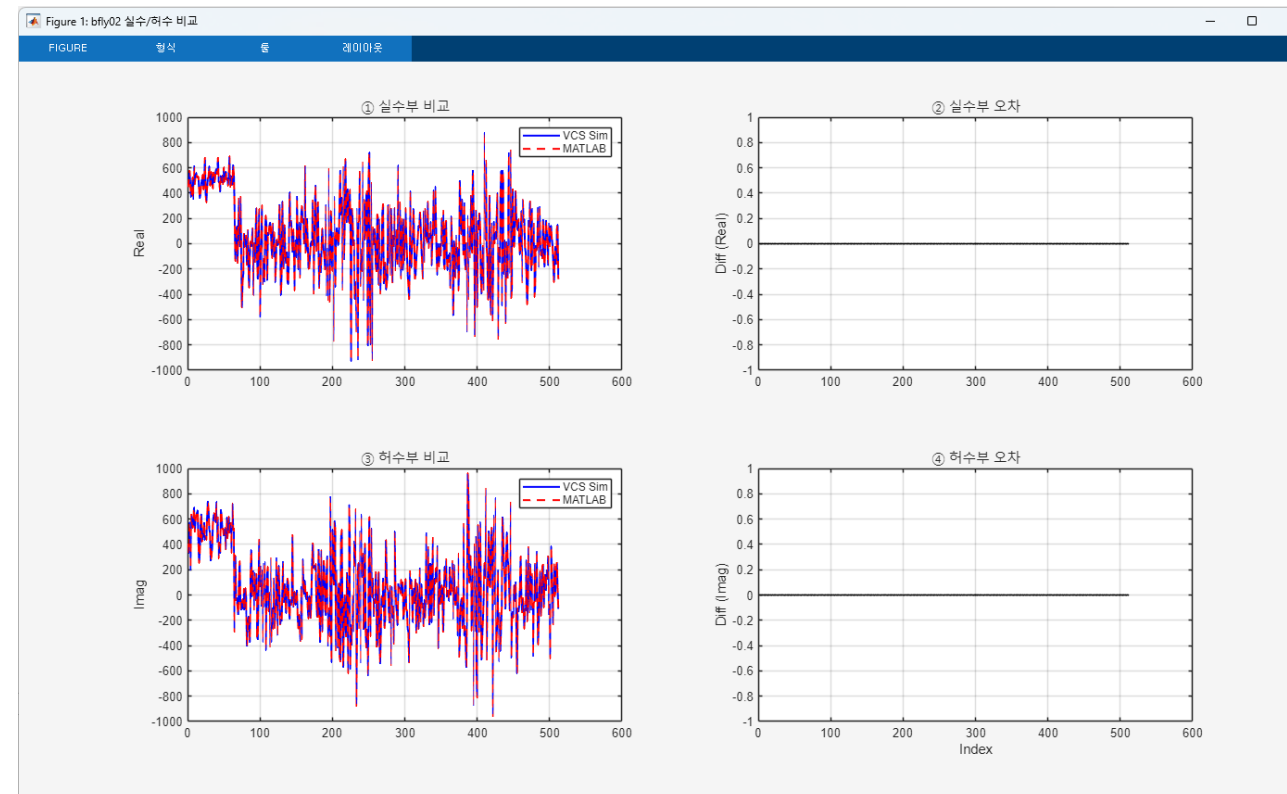
Module0 – step0_1

```
Ver [icon] valid_fac8_0_out
= step0_1
Ver [icon] bfly01_tmp_i[511:0][10:0]
Ver [icon] bfly01_tmp_q[511:0][10:0]
Ver [icon] valid_bfly01_tmp_out
Ver [icon] temp_bfly01_i[511:0][20:0]
Ver [icon] temp_bfly01_q[511:0][20:0]
Ver [icon] valid_temp_bfly01_out
Ver [icon] bfly01_i[511:0][11:0]
Ver [icon] bfly01_q[511:0][11:0]
Ver [icon] valid_bfly01_out
```

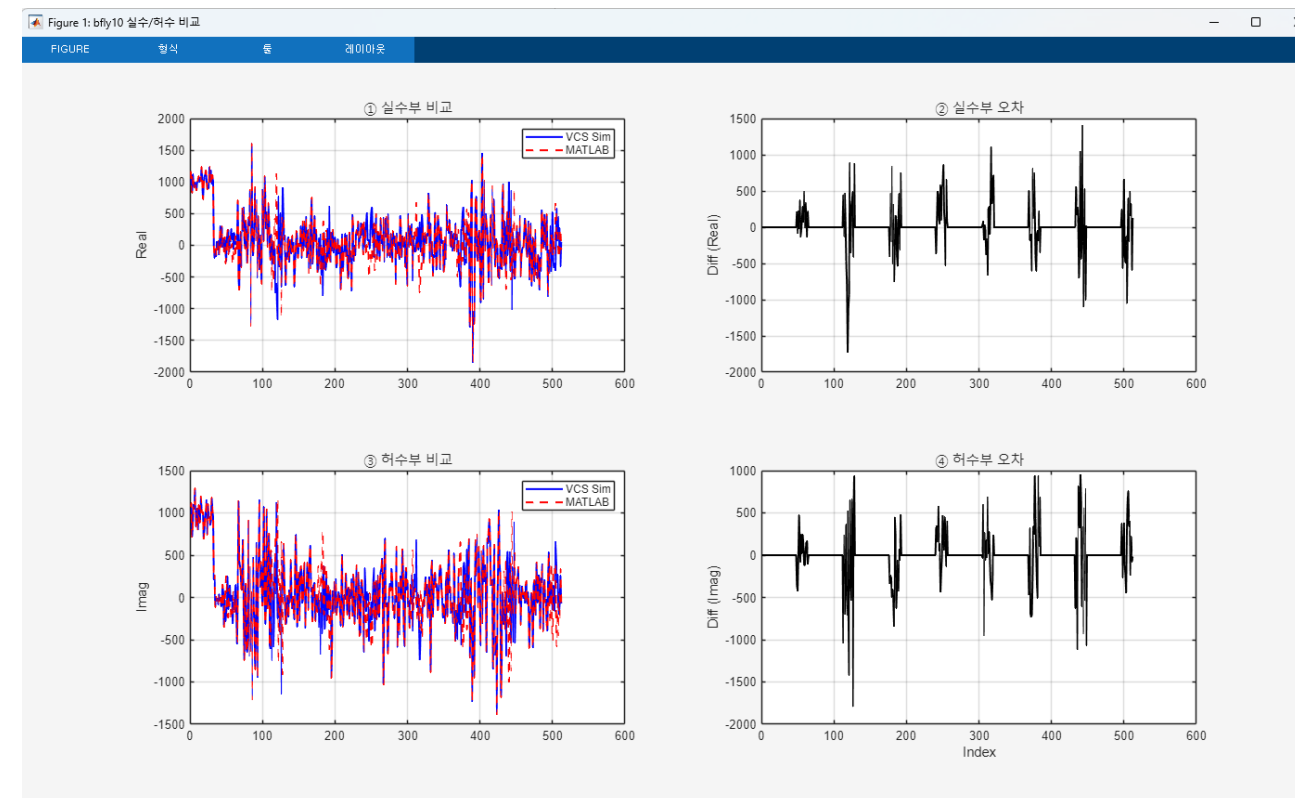
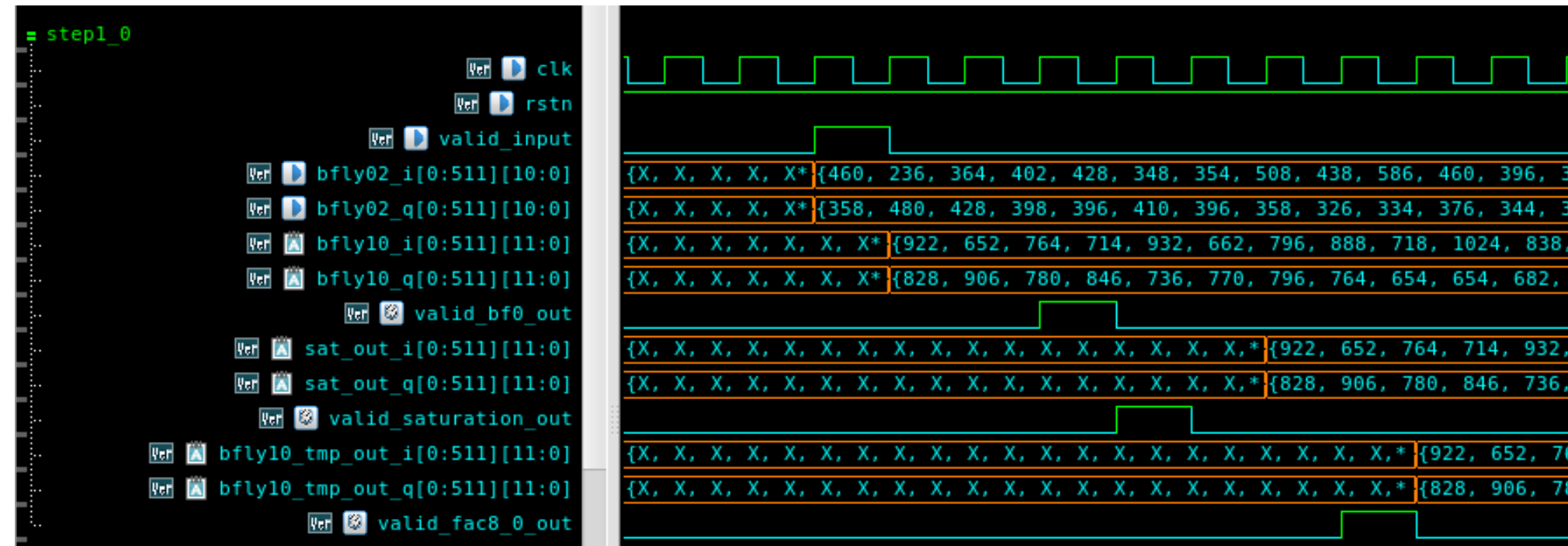
```
{-21, -33, -70, -43, -41, 21, -51, -*{-6, -2, -64, -21, -28, 42, -46, 3, -36,
{51, 24, 26, 35, 10, -5, 42, 27, -1,*{11, 2, -5, 30, -25, -34, 6, -21, -13, -
{13032, 10317, 17376, 14118, 9231, -4706, 16833, 7*{3077, 724, 10679, 9231, 5
{-5430, 1629, 7964, 1448, 5611, -2896, 1629, -2353*{-905, 0, 12489, -1629, 95
{51,*{51, 40, 68, 55, 36, -18, 66, 29, 50, 14, 45, 30, 49, 5*{12, 3, 42, 36,
{-21*{-21, 6, 31, 6, 22, -11, 6, -9, 52, 17, -8, -9, -3, 50,*{-4, 0, 49, -6,
```



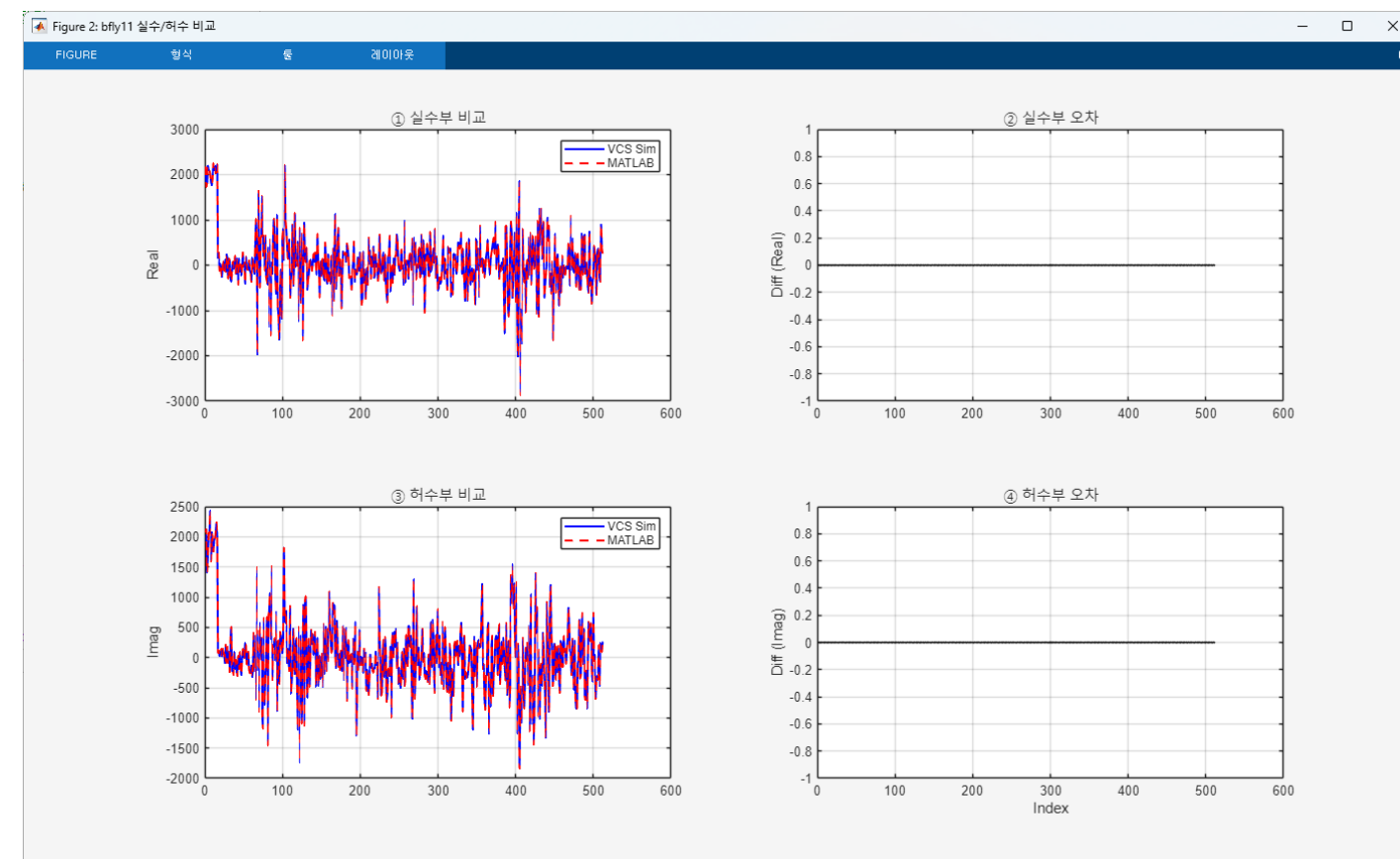
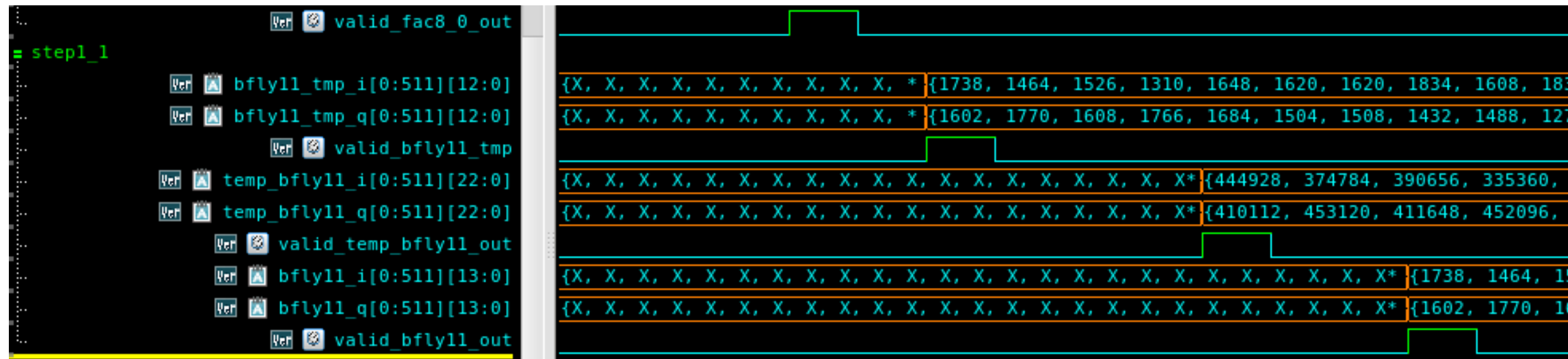
Module0 – step0_2



Module1 – step1_0



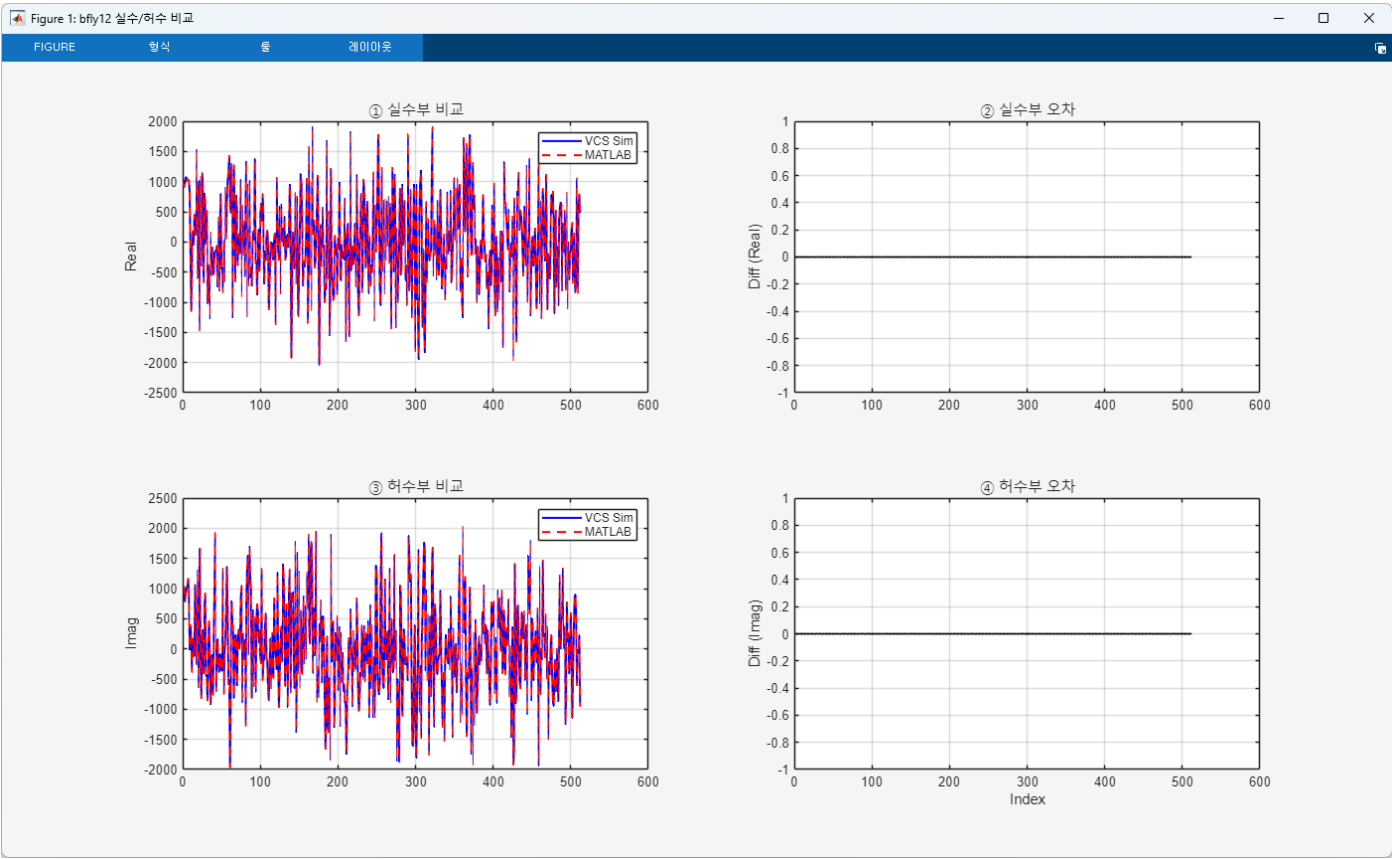
Module1 – step1_1



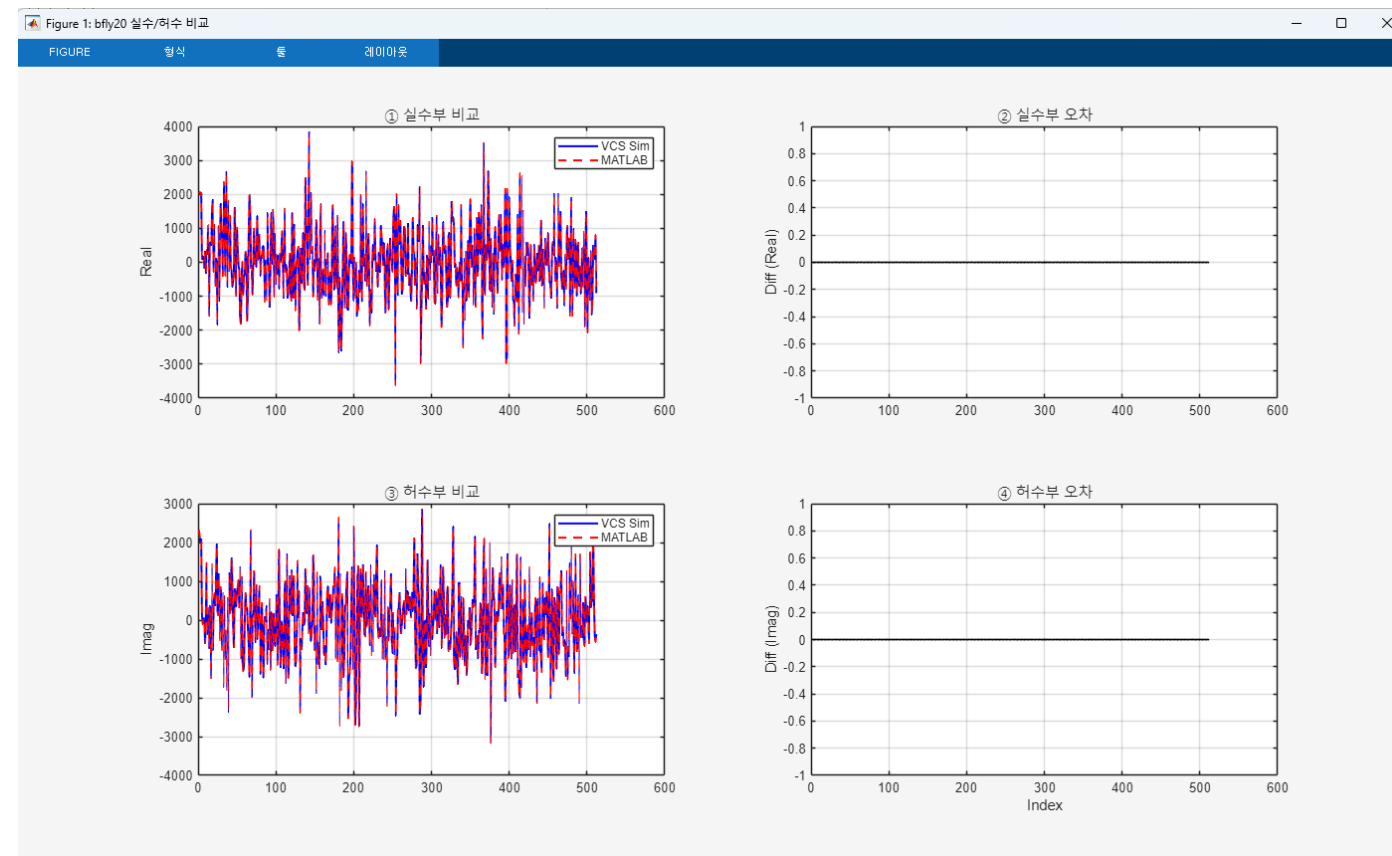
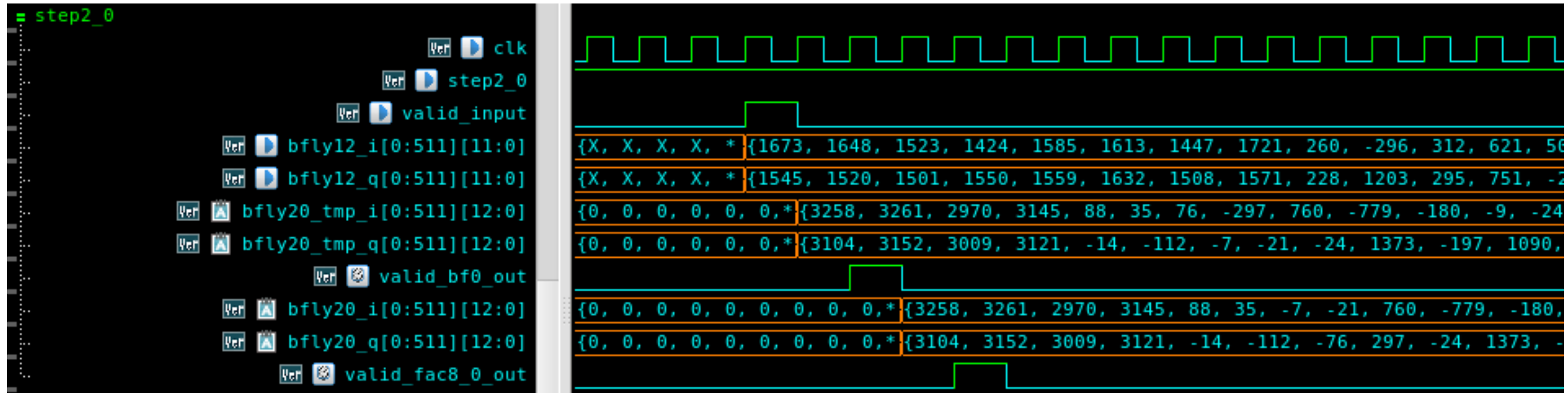
Module1 – step1_2

```
valid_bfly11_out
step1_2
  Ver A bfly12_tmp_i[0:511][14:0]
  Ver A bfly12_tmp_q[0:511][14:0]
  Ver step1_1
  Ver A STRING FORMAT ERROR AT POSITION 1
  Ver A pre_bfly12_q[0:511][24:0]
  Ver valid_mul_twf_m1_out
  Ver A pre_bfly12_real_buf[0:511][24:0]
  Ver A pre_bfly12_imag_buf[0:511][24:0]
  Ver valid_start_out
  Ver bfly12_i[0:511][11:0]
  Ver bfly12_q[0:511][11:0]
  Ver valid_output
```

```
{X, X, X, X, X, X, X,*{3346, 3296, 3046, 2848, 3170, 3226, 2894, 3442, 130, -368, 6, -228, 126, 14, 346, 226, -74, -122, -28, 4, 226, -348
{X, X, X, X, X, X, X,*{3090, 3040, 3002, 3100, 3118, 3264, 3016, 3142, 114, 500, 214, 432, 250, -256, 0, -278, 226, -174, -204, 216, -242,
{X, X, X, X, X, X, X,*{428288, 421888, 389888, 364544, 405760, 412928, 370432, 440576, 16640, -18924, 20020, 39804, 32000, -30894, -314
{X, X, X, X, X, X, X,*{395520, 389120, 384256, 396800, 399104, 417792, 386048, 402176, 14592, 77032, 18928, 48072, -16128, 10892, -3148
{0, 0, 0, 0, 0, 0, 0, 0,*{428288, 421888, 389888, 364544, 405760, 412928, 370432, 440576, 16640, -18924, 20020, 39804, 32000, -30894, -31
{0, 0, 0, 0, 0, 0, 0, 0,*{395520, 389120, 384256, 396800, 399104, 417792, 386048, 402176, 14592, 77032, 18928, 48072, -16128, 10892, -314
{X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X,*{1673, 1648,
{X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X, X,*{1545, 1520,
```



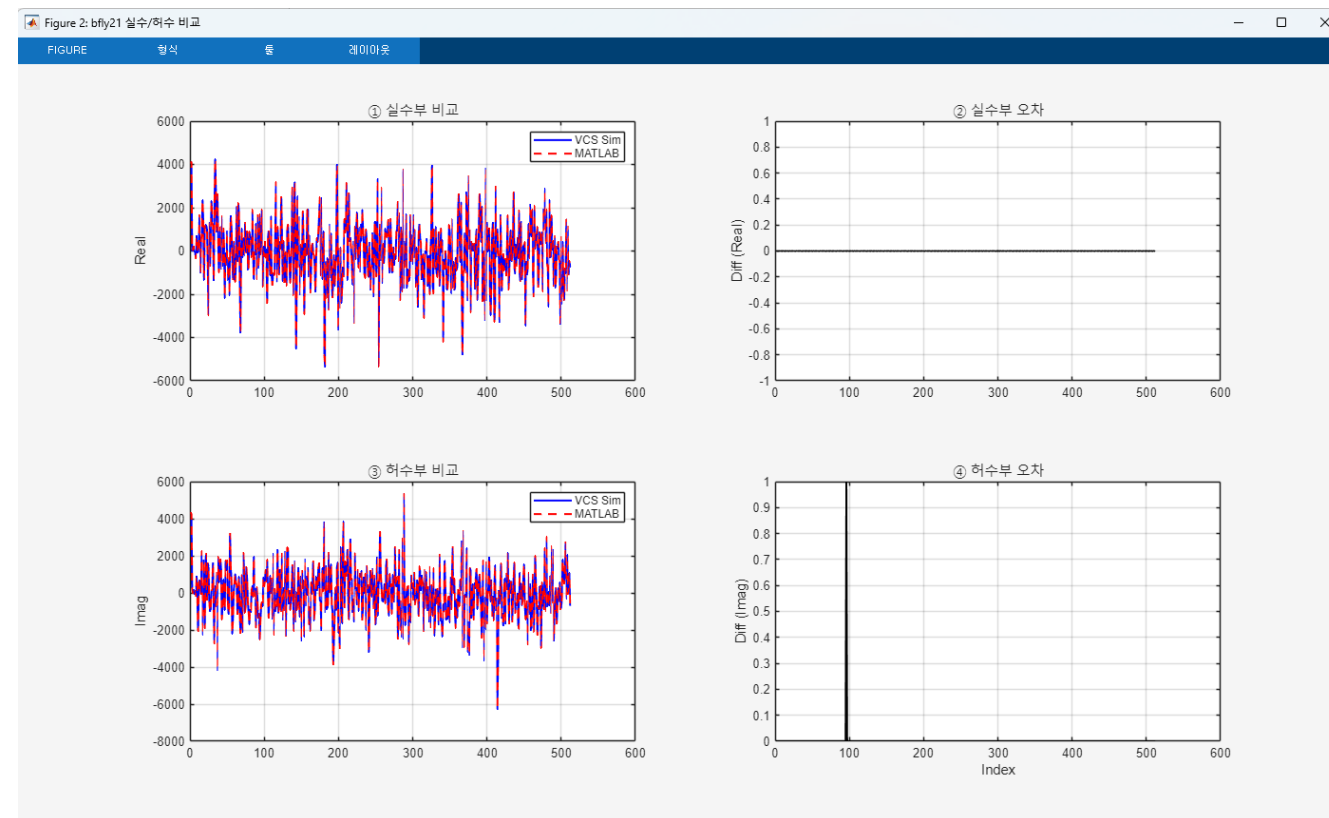
Module2 – step2_0



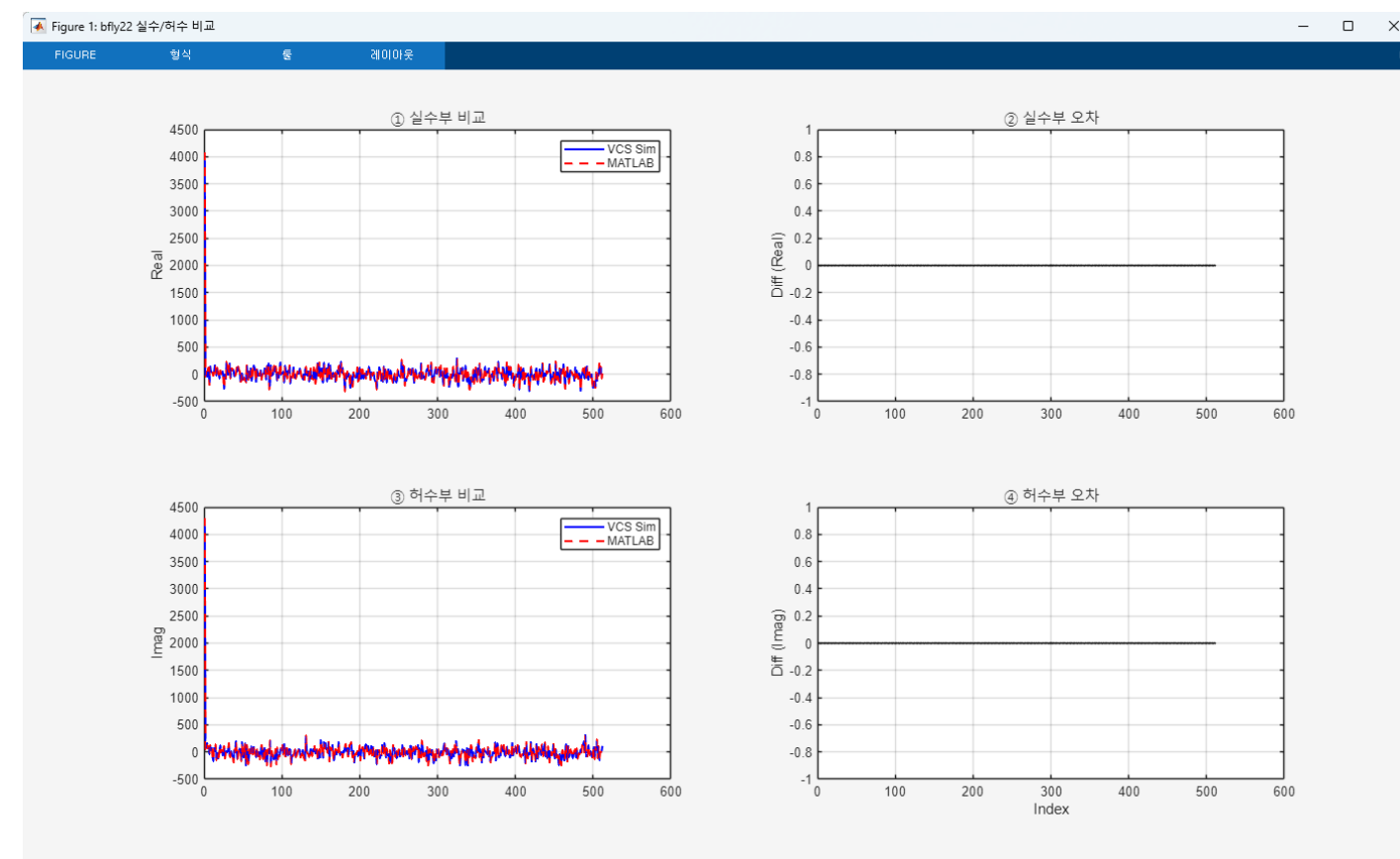
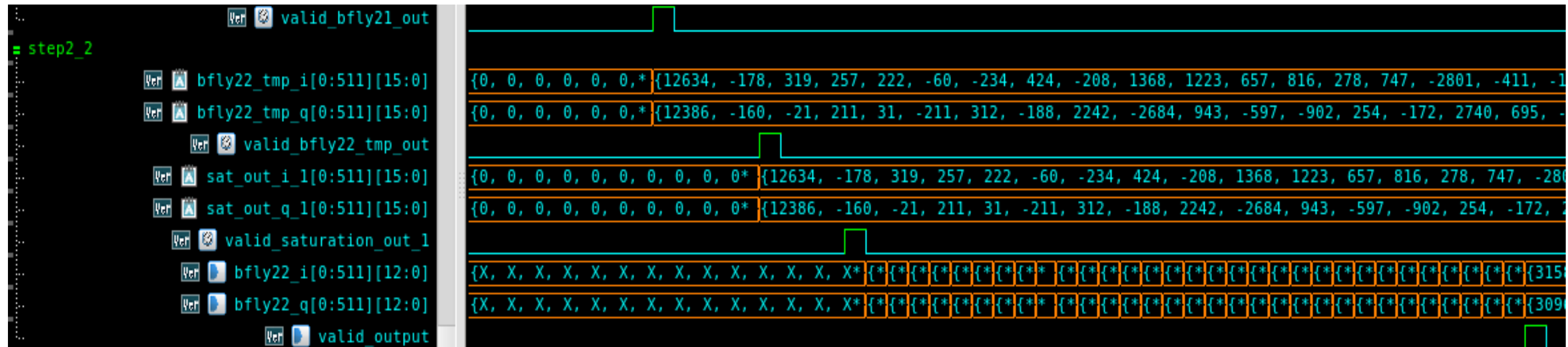
Module2 – step2_1

```
Ver valid_fac8_0_out
= step2_1
Ver bfly21_tmp_i[511:0][13:0]
Ver bfly21_tmp_q[511:0][13:0]
Ver valid_bfly01_tmp_out
Ver sat_out_i[0:511][13:0]
Ver sat_out_q[0:511][13:0]
Ver valid_saturation_out
Ver temp_bfly21_i[0:511][23:0]
Ver temp_bfly21_q[0:511][23:0]
Ver valid_temp_bfly21_out
Ver bfly21_i[0:511][14:0]
Ver bfly21_q[0:511][14:0]
Ver valid_bfly21_out
```

Waveform visualization showing digital signals and numerical data arrays. The signals are represented by green and red step functions. The numerical data is shown as arrays of integers, such as {0, 0, 0, 0, *} and {-223, 335, 499, -937, 2514, 1367, 1702, 351, -2855, 919, -683, -1607, -1051, 0}.



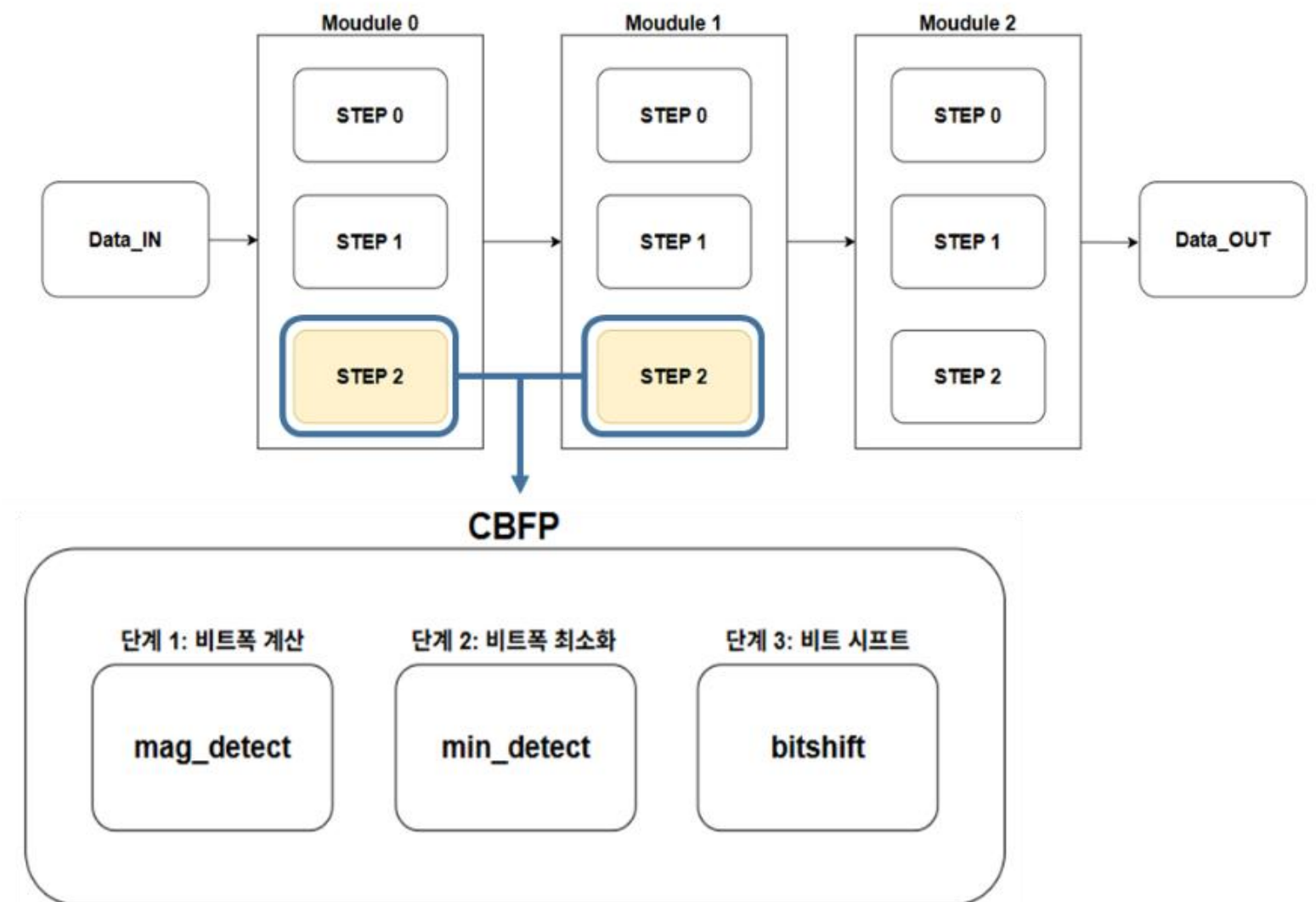
Module2 – step2_2





PART 2. CBFP Simulation & Verification

CBFP

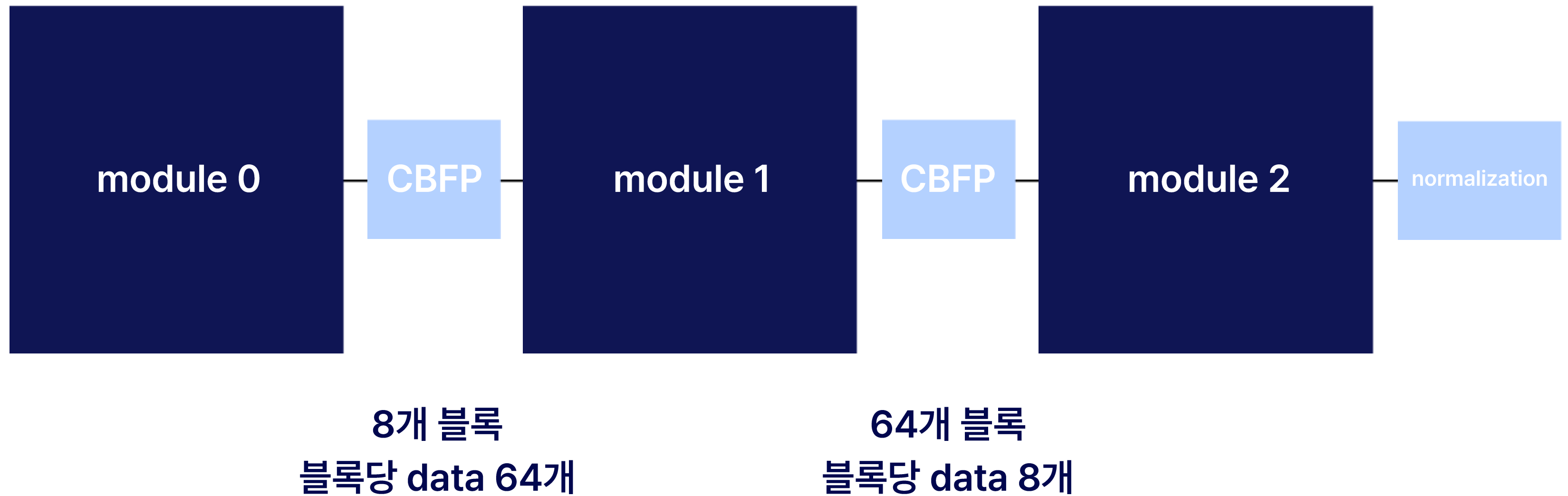


목적: 고정소수점 연산에서 공동 스케일링을 적용해 동적 범위 확장 및 오버플로 방지.

방식: 블록 단위로 입력 데이터의 최소 Headroom 탐색 → Shift 스케일링 적용

장점: 정밀도 유지 + 하드웨어 효율성 확보, FFT 등 대규모 연산에서 메모리와 연산자 절약

CBFP



CBFP

MSB == 1

```
if ((d
case.
23'b111111111111111111111111: headroom <= 22;
23'b111111111111111111111111?: headroom <= 21;
23'b111111111111111111111111?: headroom <= 20;
23'b111111111111111111111111??? headroom <= 19;
23'b111111111111111111111111????: headroom <= 18;
23'b111111111111111111111111????: headroom <= 17;
23'b111111111111111111111111????: headroom <= 16;
23'b111111111111111111111111????: headroom <= 15;
23'b111111111111111111111111????: headroom <= 14;
23'b111111111111111111111111????: headroom <= 13;
23'b111111111111111111111111????: headroom <= 12;
23'b111111111111111111111111????: headroom <= 11;
23'b111111111111111111111111????: headroom <= 10;
23'b111111111111111111111111????: headroom <= 9;
23'b111111111111111111111111????: headroom <= 8;
23'b111111111111111111111111????: headroom <= 7;
23'b111111111111111111111111????: headroom <= 6;
23'b111111111111111111111111????: headroom <= 5;
23'b111111111111111111111111????: headroom <= 4;
23'b111111111111111111111111????: headroom <= 3;
23'b111111111111111111111111????: headroom <= 2;
23'b111111111111111111111111????: headroom <= 1;
23'b111111111111111111111111????: headroom <= 0;
```

MSB == 0

```
end
ov
end
els
case.
23'b000000000000000000000000: headroom <= 22;
23'b000000000000000000000000?: headroom <= 21;
23'b000000000000000000000000?: headroom <= 20;
23'b000000000000000000000000??? headroom <= 19;
23'b000000000000000000000000????: headroom <= 18;
23'b000000000000000000000000????: headroom <= 17;
23'b000000000000000000000000????: headroom <= 16;
23'b000000000000000000000000????: headroom <= 15;
23'b000000000000000000000000????: headroom <= 14;
23'b000000000000000000000000????: headroom <= 13;
23'b000000000000000000000000????: headroom <= 12;
23'b000000000000000000000000????: headroom <= 11;
23'b000000000000000000000000????: headroom <= 10;
23'b000000000000000000000000????: headroom <= 9;
23'b000000000000000000000000????: headroom <= 8;
23'b000000000000000000000000????: headroom <= 7;
23'b000000000000000000000000????: headroom <= 6;
23'b000000000000000000000000????: headroom <= 5;
23'b000000000000000000000000????: headroom <= 4;
23'b000000000000000000000000????: headroom <= 3;
23'b000000000000000000000000????: headroom <= 2;
23'b000000000000000000000000????: headroom <= 1;
23'b000000000000000000000000????: headroom <= 0;
default: headroom <= 23;
endcase
over <= 1;
```

구현내용



과정	설명	구현 방법
mag_detect	headroom을 찾는 단계	MUX(case)
min_detect	최솟값을 찾는 단계	Combinational Logic
shift	스케일링 단계	shift 연산자(<<< , >>>)

CBFP

BLOCK



12bit

Decimal	Binary	headroom
-20	1111 1110 1100	6
-10	1111 1111 0110	7
0	0000 0000 0000	11
10	0000 0000 1010	7
20	0000 0001 0100	6
30	0000 0001 1110	6
40	0000 0010 1000	5
50	0000 0011 0010	5



CBFP



7bit

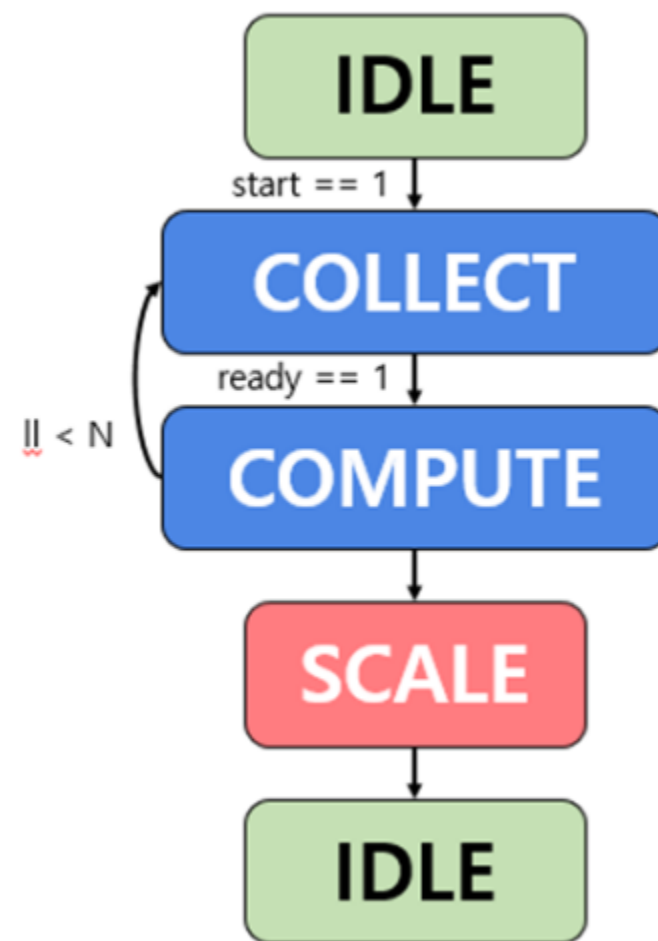
Binary	
1	10 1100
1	11 0110
0	00 0000
0	00 1010
0	01 0100
0	01 1110
0	10 1000
0	11 0010

CBFP02

23bit를 11bit로 스케일링 하기

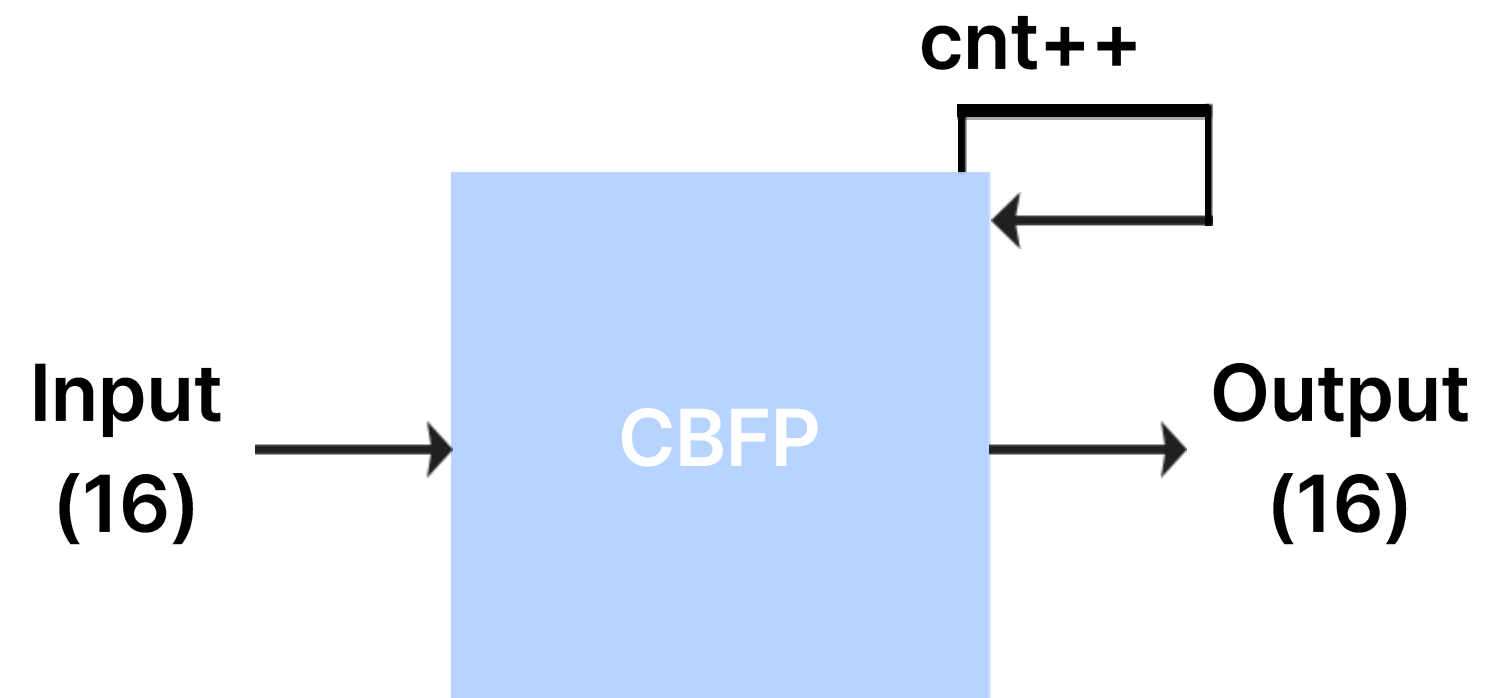
CBFP12

25bit를 12bit로 스케일링 하기



512 / FSM, CNT

V1

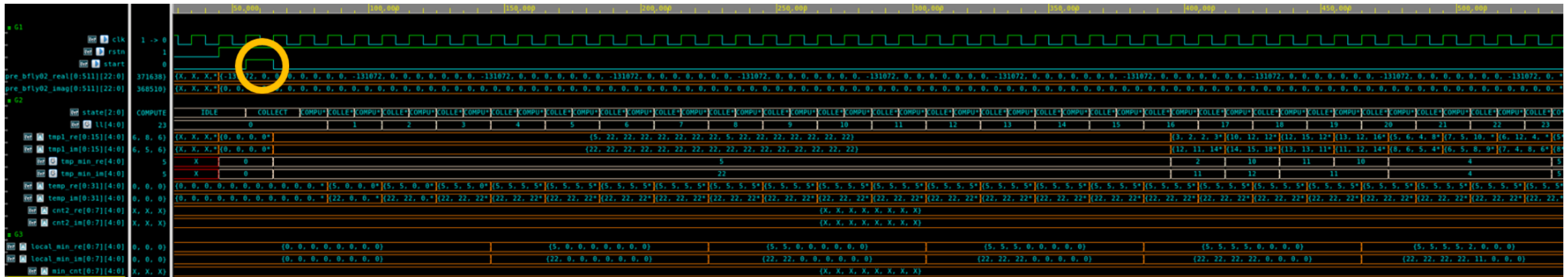


16 / CNT

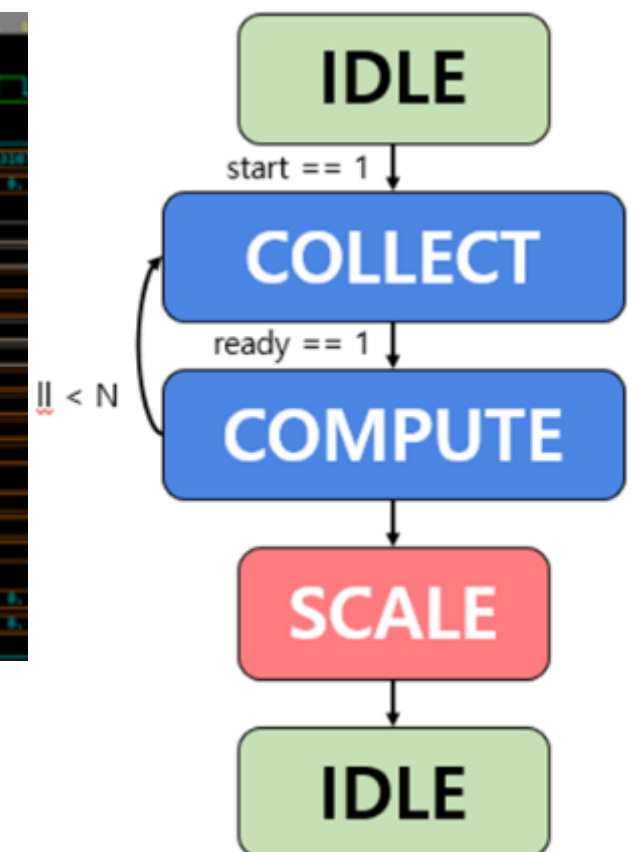
V2

512 / FSM, CNT → Input: cosine(pre_bfly02)

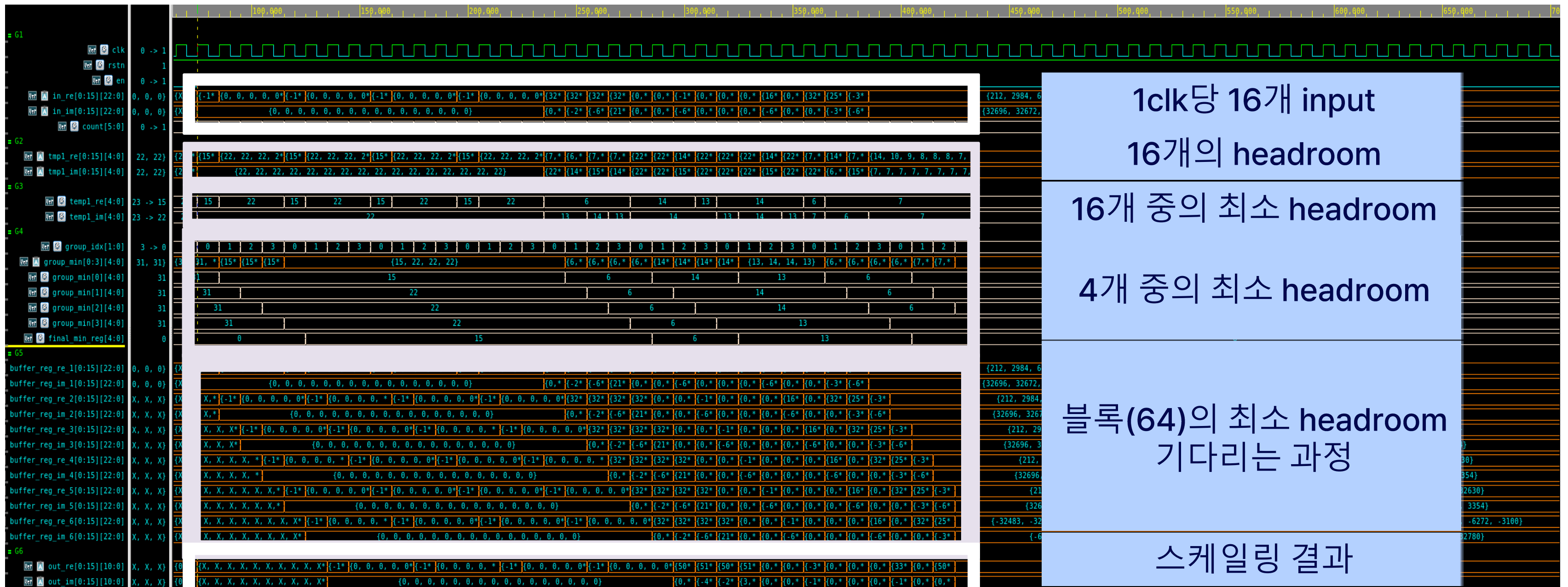
16개씩 처리



Latency: 68clk

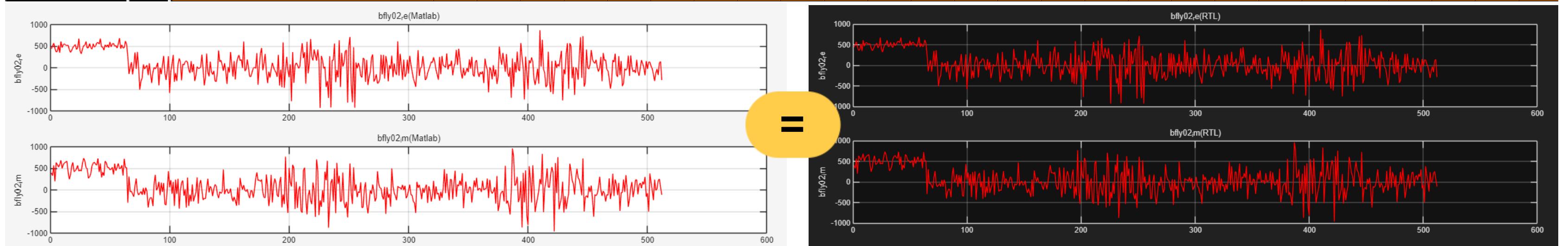
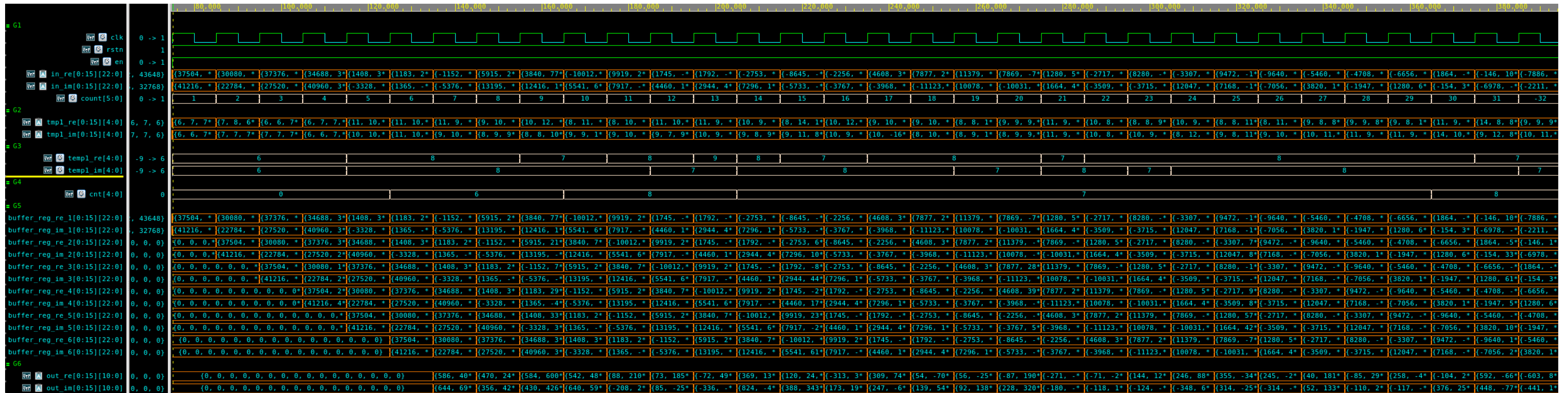


16 / CNT → Input: cosine(pre_bfly02)



Latency: 38clk

16 / CNT → Input: random(pre_bfly02)



CBFP02 AREA

Report : timing		
-path full	0.00	0.00
-delay max		
-max_paths 1		
Design : CBFP_02		
Version: V-2023.12-SP5-4		
Date : Mon Aug 4 17:54:25 2025		

# A fanout number of 1000 was used for high fanout net computations.		
Operating Conditions: TT_0P80V_0P00V_0P00V_25C Library: GF22FDX_SC7P5T		
Wire Load Model Mode: enclosed		
Startpoint: GEN_HEADROOM_0_u_hd_in/headroom_reg_3		
(rising edge-triggered flip-flop clocked by cnt_clk)		
Endpoint: temp_in_reg_1_2		
(rising edge-triggered flip-flop clocked by cnt_clk)		
Path Group: cnt_clk		
Path Type: max		
Point	Incr	Path
clock cnt_clk (rise edge)	0.00	0.00
clock network delay (ideal)	0.00	0.00
GEN_HEADROOM_0_u_hd_in/headroom_reg_3_CLK (SC7P5T_SOFFQK4_CSC20L)	0.00	0.00 r
GEN_HEADROOM_0_u_hd_in/headroom_reg_3_Q (SC7P5T_SOFFQK4_CSC20L)	43.03	43.03 f
GEN_HEADROOM_0_u_hd_in/headroom[3] (headroom_detect_23_30)	0.00	43.03 f
U411207/Z (SC7P5T_MK21AK2_CSC20L)	21.41	64.44 f
U258974/Z (SC7P5T_INW3_CSC20L)	0.38	72.82 r
U258973/Z (SC7P5T_OAI21X1_CSC20L)	11.15	83.97 f
U247066/Z (SC7P5T_OAI21X1_CSC20L)	26.15	110.12 r
U411316/Z (SC7P5T_MK21AK2_CSC20L)	22.74	132.85 f
U411327/Z (SC7P5T_MK21AK2_CSC20L)	34.19	167.05 f
U411330/Z (SC7P5T_OAI21X2_CSC20L)	24.18	191.23 r
U259344/Z (SC7P5T_OAI21X2_CSC20L)	20.95	212.19 f
U411331/Z (SC7P5T_MK21AK2_CSC20L)	33.33	245.51 f
U411337/Z (SC7P5T_OAI21X1_CSC20L)	20.38	276.89 r
U253063/Z (SC7P5T_OAI21X2_CSC20L)	27.72	298.61 f
U411338/Z (SC7P5T_MK21AK2_CSC20L)	15.19	313.80 r
U247504/Z (SC7P5T_AOI21X2_CSC20L)	14.06	327.86 f
U411342/Z (SC7P5T_AOI21X4_CSC20L)	24.58	352.44 r
U411346/Z (SC7P5T_MK21AK2_CSC20L)	12.00	364.53 f
U411347/Z (SC7P5T_AOI21X3_CSC20L)	14.75	379.28 r
U411352/Z (SC7P5T_OAI22X2_CSC20L)	19.49	398.77 f
U411353/Z (SC7P5T_OAI21X4_CSC20L)	20.63	419.40 r
U240032/Z (SC7P5T_INW3_CSC20L)	11.11	430.51 f
U240030/Z (SC7P5T_INW2_CSC20L)	7.62	438.13 r
U250455/Z (SC7P5T_AOI21X2_CSC20L)	12.43	450.56 f
U254104/Z (SC7P5T_OAI22X2_CSC20L)	15.57	466.13 r
U251940/Z (SC7P5T_OAI21X3_CSC20L)	20.20	486.33 f
U247050/Z (SC7P5T_INW3_CSC20L)	14.11	500.44 r
U251939/Z (SC7P5T_MK21AK2_CSC20L)	0.38	509.42 f
U411363/Z (SC7P5T_AOI21X3_CSC20L)	11.94	521.37 r
U259548/Z (SC7P5T_MK21AK2_CSC20L)	10.53	531.89 f
U259440/Z (SC7P5T_OAI21X2_CSC20L)	18.93	550.82 r
U411368/Z (SC7P5T_MK21AK2_CSC20L)	29.91	580.73 f
U411372/Z (SC7P5T_MK21AK2_CSC20L)	21.46	604.21 r
U411373/Z (SC7P5T_AOI22X2_CSC20L)	25.59	629.80 f
U411374/Z (SC7P5T_INW2_CSC20L)	12.81	642.61 f
U259550/Z (SC7P5T_OAI21X1_CSC20L)	17.79	660.40 r
U254098/Z (SC7P5T_OAI21X2_CSC20L)	19.43	679.83 f
U411392/Z (SC7P5T_MK21AK2_CSC20L)	33.80	713.64 f
U411397/Z (SC7P5T_AOI21X1_CSC20L)	27.50	741.14 r
U411908/Z (SC7P5T_MK21AK2_CSC20L)	19.51	760.64 f
U412000/Z (SC7P5T_MK21AK2_CSC20L)	32.99	793.64 f
U412005/Z (SC7P5T_OAI21X1_CSC20L)	23.75	817.39 r
U259552/Z (SC7P5T_MK21AK2_CSC20L)	17.35	834.73 f
U259401/Z (SC7P5T_MK21AK2_CSC20L)	30.20	871.02 r
U259402/Z (SC7P5T_INW2_CSC20L)	9.20	882.22 f
U250200/Z (SC7P5T_OAI21X3_CSC20L)	27.59	909.81 f
U259432/Z (SC7P5T_MK21AK2_CSC20L)	27.01	936.82 f
U259695/Z (SC7P5T_AOI21X1_MK_CSC20L)	23.78	960.60 r
U412017/Z (SC7P5T_OAI21X2_CSC20L)	22.38	982.98 f
U254106/Z (SC7P5T_MK21AK2_CSC20L)	40.10	1023.08 r
U254105/Z (SC7P5T_INW3_CSC20L)	9.55	1033.03 f
U412024/Z (SC7P5T_OAI21X3_CSC20L)	17.49	1050.52 r
U412035/Z (SC7P5T_OAI21X4_CSC20L)	17.28	1067.80 f
U412036/Z (SC7P5T_MK21AK2_CSC20L)	32.40	1100.20 f
U412031/Z (SC7P5T_OAI21X1_CSC20L)	21.70	1121.99 r
U259400/Z (SC7P5T_OAI21X2_CSC20L)	30.12	1152.11 r
U251913/Z (SC7P5T_INW3_CSC20L)	10.47	1162.58 f
U247400/Z (SC7P5T_MK21AK2_CSC20L)	30.78	1193.35 f
U254099/Z (SC7P5T_AOI21X2_CSC20L)	24.47	1217.82 r
U253076/Z (SC7P5T_OAI21X3_CSC20L)	24.70	1242.53 f
U251909/Z (SC7P5T_INW4_CSC20L)	21.43	1263.96 r
U258969/Z (SC7P5T_MK21AK2_CSC20L)	14.09	1278.04 f
U412042/Z (SC7P5T_OAI21X6_CSC20L)	22.02	1300.06 r
temp_in_reg_1_2_0 (SC7P5T_SOFFQK4_CSC20L)	0.00	1300.06 r
data arrival time		
clock cnt_clk (rise edge)	1400.00	1400.00
clock network delay (ideal)	0.00	1400.00
clock uncertainty	-50.00	1350.00 r
temp_in_reg_1_2_0_CLK (SC7P5T_SOFFQK4_CSC20L)	0.00	1350.00 r
library setup time	-49.82	1300.18
data required time		
data required time		1300.18
data arrival time		
data arrival time		-1300.06
slack (MET)		
slack (MET)		0.12

GF22FDX_SC7P5T_116CPP_BASE_CSC20L_TT_0P80V_0P00V_0P00V_0P00V_PP_BASE_CSC20L_TT_0P80V_0P00V_0P00V_0P00V_25C.db)

Number of ports:	41236
Number of nets:	241535
Number of cells:	216783
Number of combinational cells:	204709
Number of sequential cells:	11974
Number of macros/black boxes:	0
Number of buf/inv:	48747
Number of references:	270

Combinational area:	64211.568428
Buf/Inv area:	8959.956063
Noncombinational area:	19120.442858
Macro/Black Box area:	0.000000
Net Interconnect area:	undefined (No wire load specified)
Total cell area:	83332.011285
Total area:	undefined

GF22FDX_SC7P5T_116CPP_BASE_CSC20L_TT_0P80V_0P00V_0P00V_0P00V_PP_BASE_CSC20L_TT_0P80V_0P00V_0P00V_0P00V_25C.db)

Number of ports:	2272
Number of nets:	15898
Number of cells:	14085
Number of combinational cells:	9013
Number of sequential cells:	5002
Number of macros/black boxes:	0
Number of buf/inv:	2296
Number of references:	182

Combinational area:	3681.561600
Buf/Inv area:	938.625587
Noncombinational area:	9503.879788
Macro/Black Box area:	0.000000
Net Interconnect area:	undefined (No wire load specified)
Total cell area:	13185.441388
Total area:	undefined

512 / FSM, CNT

16 / CNT

Information: Updating design information... (UID-85)		
Warning: Design 'CBFP_02' contains 3 high-fanout nets. A fanout number of 1000		

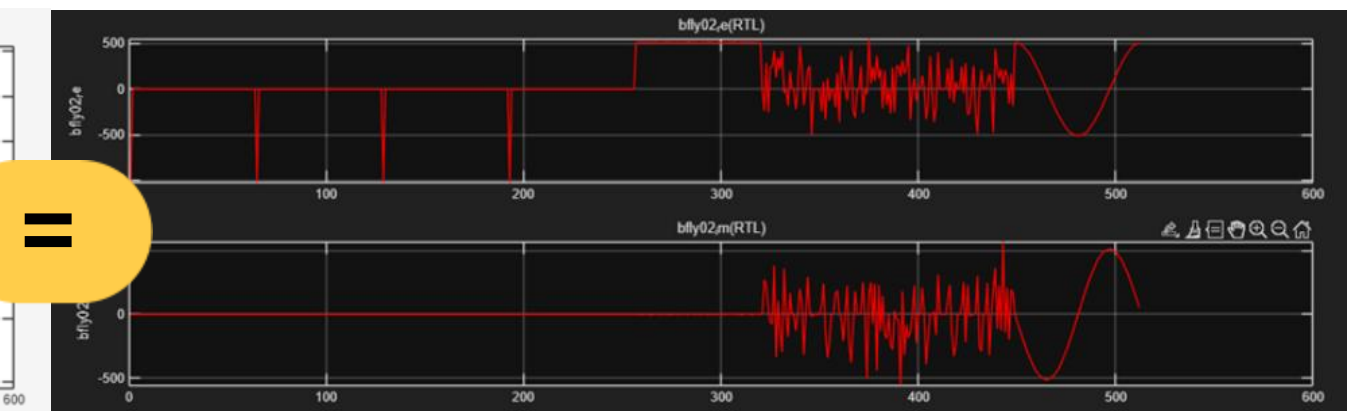
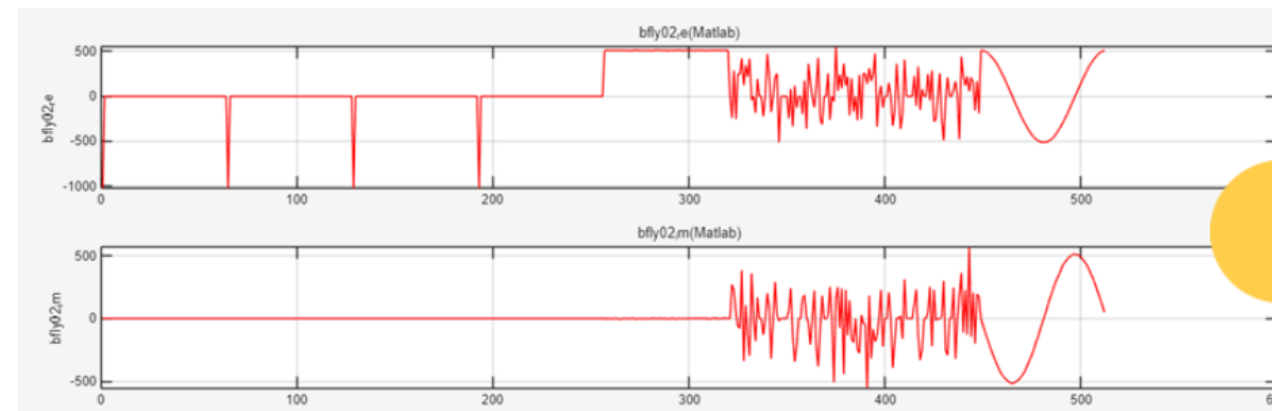
Report : timing		
-path full	0.00	0.00
-delay max		
-max_paths 1		
Design : CBFP_02		
Version: V-2023.12-SP5-4		
Date : Mon Aug 4 15:45:24 2025		

# A fanout number of 1000 was used for high fanout net computations.		
Operating Conditions: TT_0P80V_0P00V_0P00V_25C Library: GF22FDX_SC7P5T		
Wire Load Model Mode: enclosed		
Startpoint: GEN_HEADROOM_1_u_hd_in/headroom_reg_4		
(rising edge-triggered flip-flop clocked by cnt_clk)		
Endpoint: group_min_reg_0_0		
(rising edge-triggered flip-flop clocked by cnt_clk)		
Path Group: cnt_clk		
Path Type: max		
Point	Incr	Path
clock cnt_clk (rise edge)	0.00	0.00
clock network delay (ideal)	0.00	0.00
GEN_HEADROOM_1_u_hd_in/headroom_reg_4_CLK (SC7P5T_SOFFQK4_CSC20L)	0.00	0.00 r
GEN_HEADROOM_1_u_hd_in/headroom_reg_4_Q (SC7P5T_SOFFQK4_CSC20L)	55.90	55.90 f
GEN_HEADROOM_1_u_hd_in/headroom[4] (headroom_detect_21_28)	0.00	55.90 f
U6380/Z (SC7P5T_MK21AK2_CSC20L)	21.55	77.46 f
U6380/Z (SC7P5T_MK21AK2_CSC20L)	11.58	89.03 r
U7005/Z (SC7P5T_OAI21X2_CSC20L)	14.23	103.26 f
U6471/Z (SC7P5T_MK21AK2_CSC20L)	16.37	119.63 r
U6781/Z (SC7P5T_OAI21X4_CSC20L)	16.73	136.36 r
U6380/Z (SC7P5T_OAI21X3_CSC20L)	13.89	150.25 f
U6067/Z (SC7P5T_MK21AK2_CSC20L)	20.38	170.63 r
U6276/Z (SC7P5T_MK21AK2_CSC20L)	12.62	183.25 r
U6381/Z (SC7P5T_OAI21X4_CSC20L)	15.91	199.16 f
U6380/Z (SC7P5T_MK21AK2_CSC20L)	12.24	211.40 r
U6270/Z (SC7P5T_AOI21X2_CSC20L)	10.74	222.24 f
U6379/Z (SC7P5T_OAI21X4_CSC20L)	18.28	240.52 r
U6068/Z (SC7P5T_MK21AK2_CSC20L)	16.72	257.24 f
U6379/Z (SC7P5T_OAI21X3_CSC20L)	10.64	267.87 r
U6324/Z (SC7P5T_AOI21X3_MK_CSC20L)	10.90	278.78 f
U6394/Z (SC7P5T_OAI21X2_CSC20L)	19.10	297.88 r
U6391/Z (SC7P5T_AOI22X4_CSC20L)	20.07	317.94 f
U6787/Z (SC7P5T_MK21AK2_CSC20L)	16.90	334.85 r
U6780/Z (SC7P5T_AOI21X2_CSC20L)	12.14	346.98 f
U6778/Z (SC7P5T_AOI21X3_MK_CSC20L)	21.68	368.66 r
U6749/Z (SC7P5T_OAI21X3_CSC20L)	17.74	386.40 f
U6380/Z (SC7P5T_OAI21X3_CSC20L)	21.42	407.82 r
U6322/Z (SC7P5T_MK21AK2_CSC20L)	10.88	418.70 f
U6563/Z (SC7P5T_AOI21X2_CSC20L)	16.97	435.67 r
U6142/Z (SC7P5T_MK21AK2_CSC20L)	23.01	458.68 r
U6252/Z (SC7P5T_MK21AK2_CSC20L)	8.37	467.05 f
U6250/Z (SC7P5T_OAI21X4X2X2_CSC20L)	13.62	480.67 r
U6122/Z (SC7P5T_MK21AK2_CSC20L)	24.10	504.76 r
U6470/Z (SC7P5T_OAI21X4_CSC20L)	22.10	526.86 f
U6469/Z (SC7P5T_MK21AK2_CSC20L)	13.22	540.08 r
U6121/Z (SC7P5T_AOI21X3_CSC20L)	29.11	569.22 f
U6316/Z (SC7P5T_AOI22X0_CSC20L)	21.87	600.50 r
U6120/Z (SC7P5T_MK21AK2_CSC20L)	29.03	637.62 r
U6797/Z (SC7P5T_OAI21X3_CSC20L)	23.56	661.18 f
U7110/Z (SC7P5T_AOI21X2_CSC20L)	18.93	680.00 r
U6691/Z (SC7P5T_INW2_CSC20L)	12.58	692.57 f
U6692/Z (SC7P5T_MK21AK2_CSC20L)	15.23	707.80 r
U7112/Z (SC7P5T_OAI21X2_CSC20L)	20.91	728.81 f
U6312/Z (SC7P5T_MK21AK2_CSC20L)	19.53	748.34 r
U6325/Z (SC7P5T_MK21AK2_CSC20L)	12.01	760.35 f
U6324/Z (SC7P5T_OAI21X3_CSC20L)	10.26	770.61 r
U6168/Z (SC7P5T_OAI21X3_CSC20L)	20.68	791.21 f
U7113/Z (SC7P5T_AOI21X2_CSC20L)	17.40	808.61 r
U7119/Z (SC7P5T_MK21AK2_CSC20L)	17.00	825.50 f
U6491/Z (SC7P5T_MK21AK2_CSC20L)	38.29	864.19 r
U6492/Z (SC7P5T_INW2_CSC20L)	9.66	873.85 f
U7149/Z (SC7P5T_AOI21X3_CSC20L)	20.07	893.92 r
U6495/Z (SC7P5T_AOI21X2_CSC20L)	25.78	919.71 r
U6496/Z (SC7P5T_INW4_CSC20L)	10.55	930.25 f
U7158/Z (SC7P5T_MK21AK2_CSC20L)	29.90	960.16 f
U6312/Z (SC7P5T_OAI21X4X2X2_CSC20L)	25.24	985.40 r
U6310/Z (SC7P5T_MK21AK2_CSC20L)	28.53	1014.93 f
U6312/Z (SC7P5T_MK21AK2_CSC20L)	19.59	1033.62 r
U6227/Z (SC7P5T_MK21AK2_CSC20L)	11.41	1045.02 f
U6226/Z (SC7P5T_OAI21X2_CSC20L)	9.94	1054.96 f
U6222/Z (SC7P5T_MK21AK2_CSC20L)	11.57	1066.53 f
U6115/Z (SC7P5T_MK21AK2_CSC20L)	11.90	1078.43 r
U6221/Z (SC7P5T_MK21AK2_CSC20L)	15.94	1094.36 f
U6071/Z (SC7P5T_MK21AK2_CSC20L)	13.17	1107.54 r
U7140/Z (SC7P5T_OAI21X2_CSC20L)	16.50	1124.04 f
U6367/Z (SC7P5T_OAI21X3_CSC20L)	17.72	1141.75 r
U6160/Z (SC7P5T_OAI21X4X2X2_CSC20L)	21.18	1162.93 f
U6313/Z (SC7P5T_MK21AK2_CSC20L)	23.06	1185.99 r
U6365/Z (SC7P5T_MK21AK2_CSC20L)	25.40	1211.39 r
U6699/Z (SC7P5T_AOI21X2_CSC20L)	16.28	1227.67 f
U5704/Z (SC7P5T_OAI21X3_CSC20L)	24.66	1252.34 r
U5628/Z (SC7P5T_MK21AK2_CSC20L)	36.19	1288.72 f
U6691/Z (SC7P5T_INW2_CSC20L)	11.22	1299.94 r
group_min_reg_0_0_0_0 (SC7P5T_SOFFQK4_CSC20L)	0.00	1299.94 r
data arrival time		
clock cnt_clk (rise edge)	1400.00	1400.00
clock network delay (ideal)	0.00	1400.00
clock uncertainty	-50.00	1350.00 r
group_min_reg_0_0_0_0_CLK (SC7P5T_SOFFQK4_CSC20L)	0.00	1350.00 r
library setup time	-49.82	1300.18
data required time		
data required time		1300.18
data arrival time		
data arrival time		-1299.94
slack (MET)		
slack (MET)		0.24

CBFP 결과

CBFP02

bfly02 (cosine)

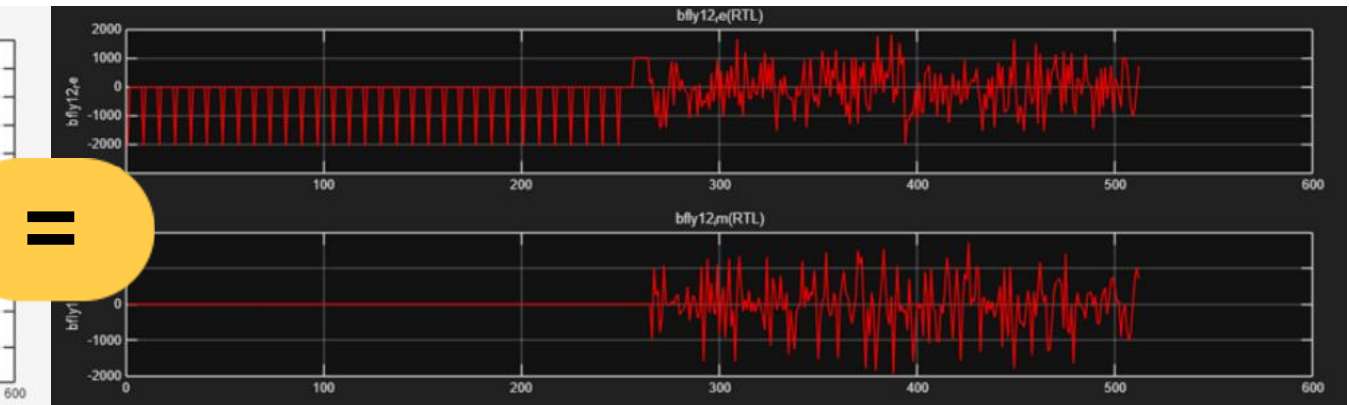
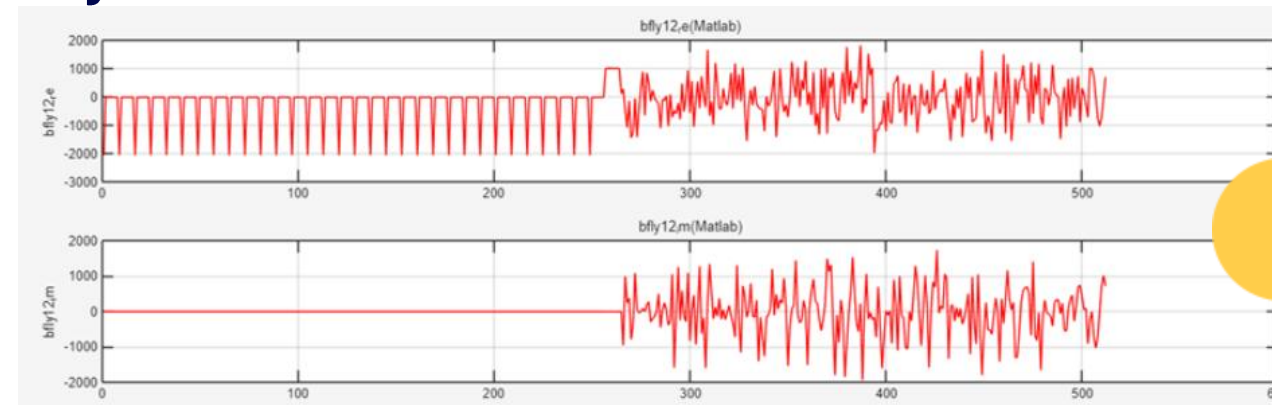


<Fixed Model>

<RTL>

CBFP12

bfly12



<Fixed Model>

<RTL>

PART 5. Synthesis



Setup Timing Check

- SQNR (Set up Timing 체크에서 slack 0.54ps로 타이밍 요구사항 만족(MET))

```
clock cnt_clk (rise edge)                1400.00    1400.00
clock network delay (ideal)                0.00    1400.00
clock uncertainty                          -50.00    1350.00
u1_module0/u1_cbfp02/temp_im_reg_0__1_/CLK (SC7P5T_SDFFRQX1_S_CSC20L)
||||| 0.00    1350.00 r
library setup time                        -53.39    1296.61
data required time                        1296.61
-----
data required time                        1296.61
data arrival time                        -1296.07
-----
slack (MET)                               0.54
```

Cell Count & Area Report

- 합성된 회로는 총 2,579,028,72의 셀 면적을 가지며, 약 264만 개의 셀로 구성

```
Library(s) Used:
  GF22FDX_SC7P5T_116CPP_BASE_CSC20L_TT_0P80V_0P00V_0P00V_0P00V_25C (Fi

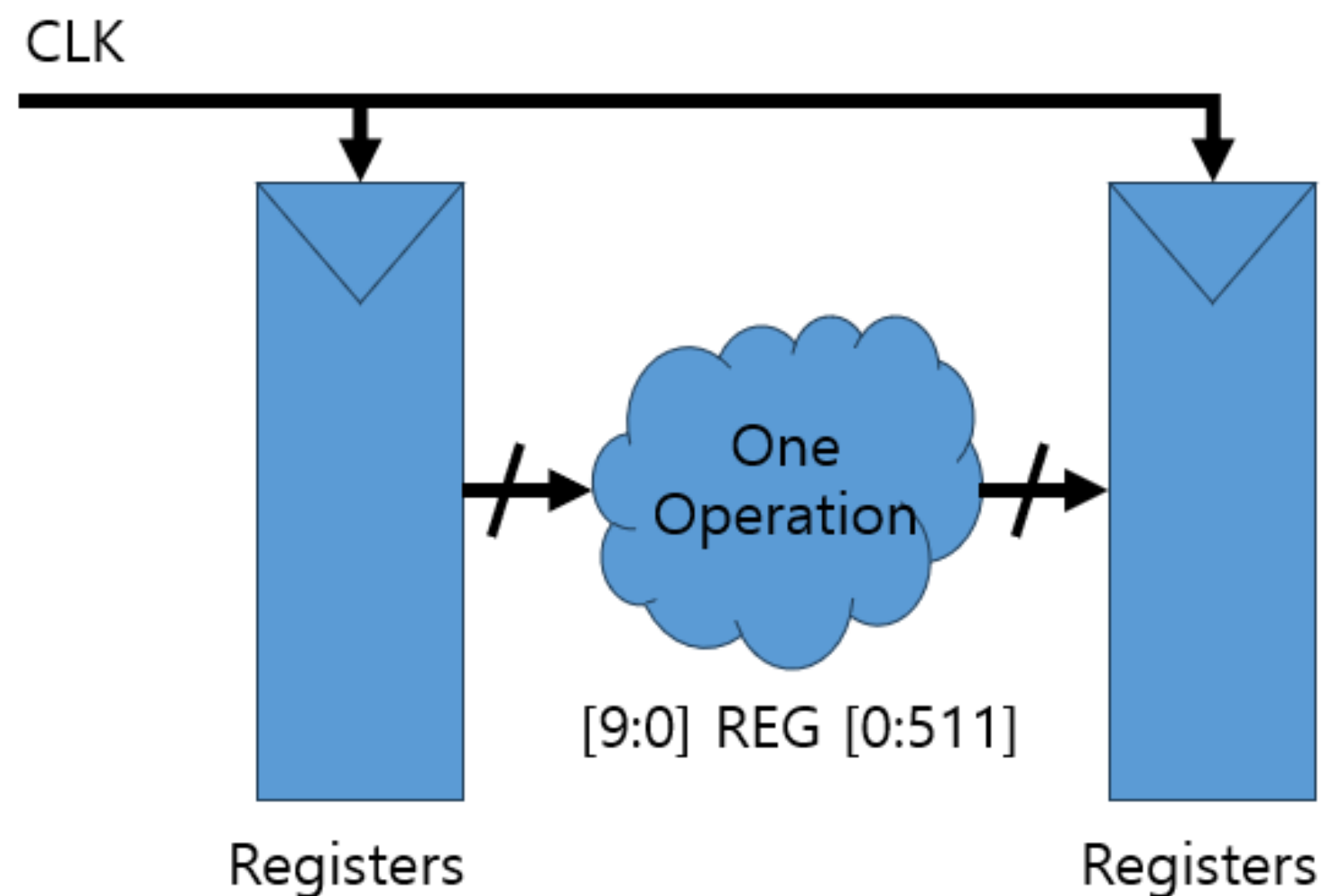
Number of ports:          1110800
Number of nets:           3617970
Number of cells:          2648318
Number of combinational cells: 1575626
Number of sequential cells: 1061152
Number of macros/black boxes: 0
Number of buf/inv:        342339
Number of references:      6

Combinational area:       656223.876991
Buf/Inv area:             56956.743001
Noncombinational area:    1922804.843500
Macro/Black Box area:     0.000000
Net Interconnect area:    undefined (No wire load specified)

Total cell area:          2579028.720491
Total area:               undefined
```

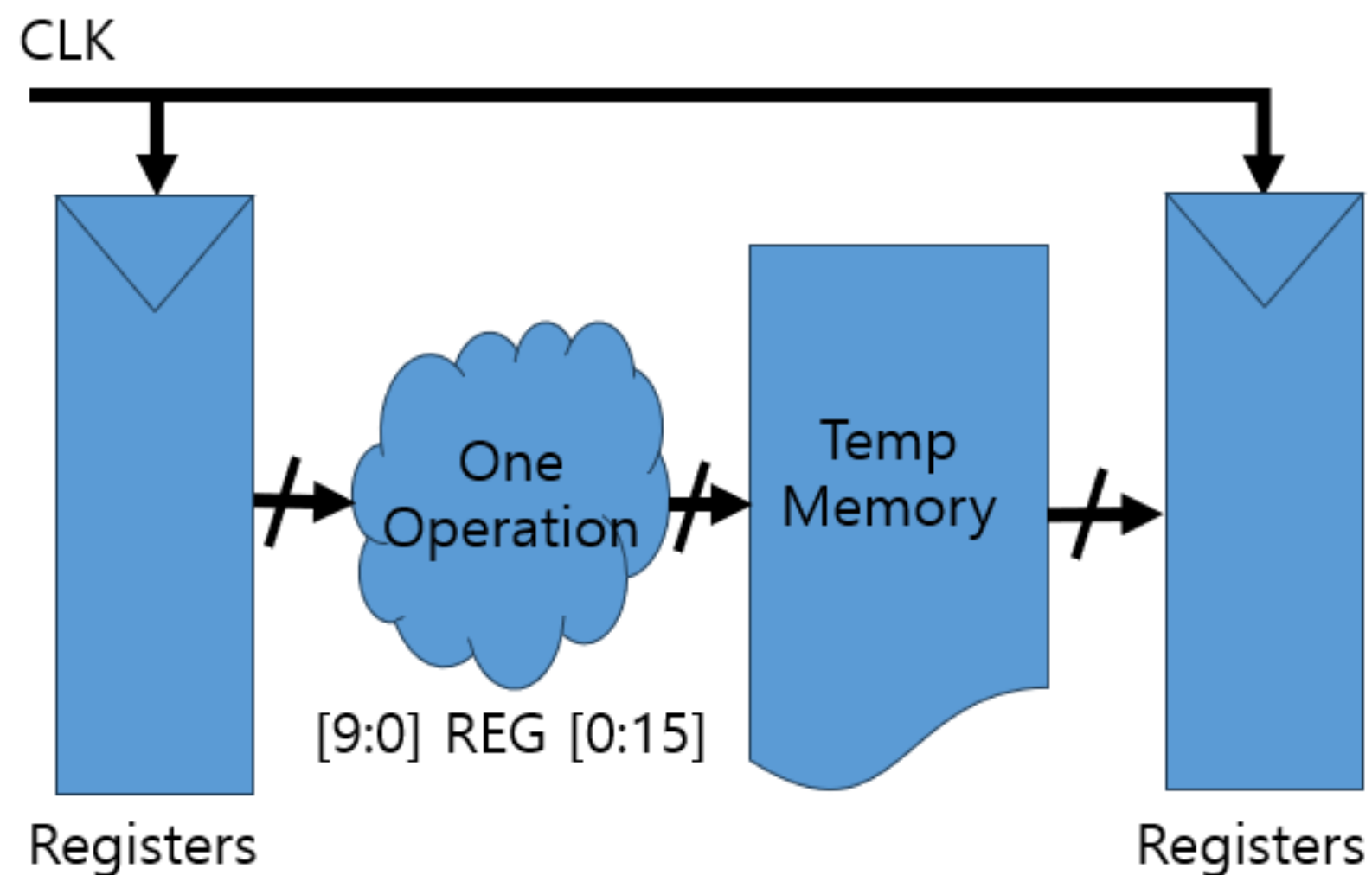
FFT Module0 간 비교

V1



Operates all 512 Points all at once

V2



Latency Increases at steps pass

FFT Module0 간 비교

	V1(512)	V2(16)	%
Area	914317	114247	87.5% 감소
LUT	338854	26673	92.1% 감소
Registers	294883	37203	87.3% 감소
Setup Time	1.52 (Slack Met)	28.66 (Slack Met)	
Latency	17	54	68% 증가

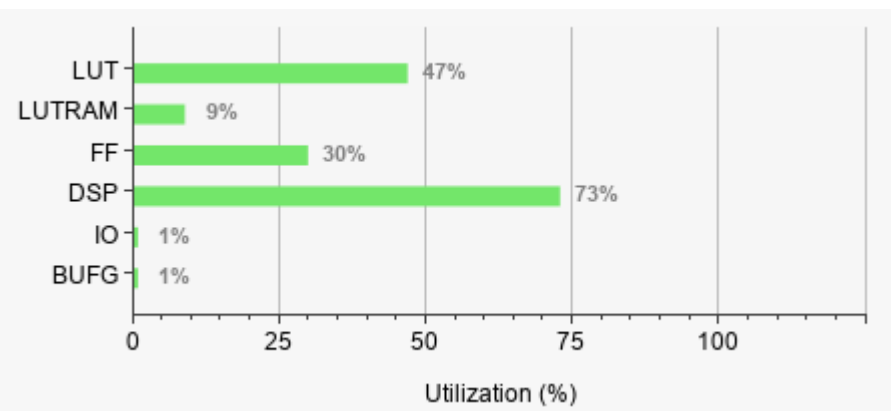
PART 6. FPGA



Synthesis 현재 상황



Alveo U200 Data Center Accelerator Card



```
# Clock constraint
create_clock -name sys_clk -period 10 [get_ports clk]

# IOSTANDARD 설정 (PACKAGE_PIN은 지정 안 함)
set_property IOSTANDARD LVCMOS18 [get_ports clk]
```

DRG (2 errors)

Pin Planning (2 errors)

- [DRG NSTD-1] Unspecified IO Standard: 1 out of 1 logical ports use IO standard (IOSTANDARD) value 'DEFAULT'. Instead of a user assigned specific value, this may cause IO contention or incompatibility with the board power or connectivity affecting performance, signal integrity or in extreme cases cause damage to the device or the components to which it is connected. To correct this violation, specify all IO standards. This design will fail to generate a bitstream unless all logical ports have a user specified IO standard value defined. To allow bitstream creation with unspecified IO standard values (not recommended), use this command: `set_property SEVERITY {Warning} [get_drc_checks NSTD-1]`. NOTE: When using the Vivado Run Infrastructure (e.g. `launch_runs Tcl` command), add this command to a .tcl file and add that file as a pre-hook for write_bitstream step for the implementation run. Problem ports: `clk`.
- [DRG UCIO-1] Unconstrained Logical Port: 1 out of 1 logical ports have no user assigned specific location constraint (LOC). This may cause IO contention or incompatibility with the board power or connectivity affecting performance, signal integrity or in extreme cases cause damage to the device or the components to which it is connected. To correct this violation, specify all pin locations. This design will fail to generate a bitstream unless all logical ports have a user specified site LOC constraint defined. To allow bitstream creation with unspecified pin locations (not recommended), use this command: `set_property SEVERITY {Warning} [get_drc_checks UCIO-1]`. NOTE: When using the Vivado Run Infrastructure (e.g. `launch_runs Tcl` command), add this command to a .tcl file and add that file as a pre-hook for write_bitstream step for the implementation run. Problem ports: `clk`.

● [Vivado 12-1345] Error(s) found during DRG. Bitgen not run.

Timing

Worst Negative Slack (WNS):	-0.548 ns
Total Negative Slack (TNS):	-600.758 ns
Number of Failing Endpoints:	3199
Total Number of Endpoints:	1705469

PART 7. Trouble Shooting



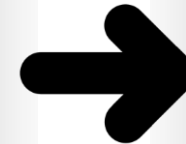
Indexsum 계산 수정

문제점

- Saturation 처리 없음
- 비트 폭이 정확하게 처리되지 않음

수정 전 코드

```
always_ff @(posedge clk or negedge rstn) begin
    if (!rstn) begin
        for (int kk = 0; kk < 512; kk++) begin
            indexsum_re_reg[kk] <= '0;
            indexsum_im_reg[kk] <= '0;
        end
    end else if (valid_in) begin
        for (int kk = 0; kk < 512; kk++) begin
            indexsum_re_reg[kk] = index1_re[kk] + index2_re[kk];
            indexsum_im_reg[kk] = index1_im[kk] + index2_im[kk];
        end
    end
end
```



수정

- 조합 논리 변경
- Saturation 처리 추가

수정된 코드

```
always_comb begin
    for (int kk = 0; kk < 512; kk++) begin
        indexsum_re_reg[kk] = index1_re[kk] + index2_re[kk];
        indexsum_im_reg[kk] = index1_im[kk] + index2_im[kk];
        // Saturate if the result exceeds 6-bit range
        if (indexsum_re_reg[kk] > 6'h3F) begin
            indexsum_re_reg[kk] = 6'h3F;
        end
        if (indexsum_im_reg[kk] > 6'h3F) begin
            indexsum_im_reg[kk] = 6'h3F;
        end
    end
end
```

Shift 연산 음수 값 처리

문제점

- Shift 연산에서 음수 값 처리 부분에서 문제 발생
- Bit shift 연산을 양수만 처리하는 알고리즘 설계

```
//
shift_val_re[i] = 9 - indexsum_re[i];
if (indexsum_re[i] >= 6'd23)
    re_bfly22_pipe[i] <= '0;
else if (shift_val_re[i] > 0)
    re_bfly22_pipe[i] <= bfly22_tmp_re[i] <<< shift_val_re[i];
else
    re_bfly22_pipe[i] <= bfly22_tmp_re[i];
```



수정

- Shift 할 값이 양수일 경우 왼쪽 shift 연산을 하고, 음수일 경우 오른쪽 shift 연산을 수행하도록 수정

```
//
shift_val_re[i] = 9 - indexsum_re[i];
if (indexsum_re[i] >= 6'd23)
    re_bfly22_pipe[i] <= '0;
else if (shift_val_re[i] > 0)
    re_bfly22_pipe[i] <= bfly22_tmp_re[i] <<< shift_val_re[i];
else if (shift_val_re[i] < 0)
    re_bfly22_pipe[i] <= bfly22_tmp_re[i] >>> -shift_val_re[i];
else
    re_bfly22_pipe[i] <= bfly22_tmp_re[i];
```

Shift 연산 음수 값 처리

문제

- 상수로 저장할 값을 logic 타입으로 선언하여 합성 중 최적화 과정에서 꼭 필요한 레지스터가 제거되는 현상 발생
=> 이로 인해 합성이 모두 진행되었음에도 불구하고, 합성 결과는 비어있게 됨.

```
logic signed [COEFF_WIDTH-1:0] coeffs [0:TAP_NUM-1] = '{  
    16'sd0, -16'sd1, 16'sd1, 16'sd0, -16'sd1, 16'sd2, 16'sd0, -16'sd2,  
    16'sd2, 16'sd0, -16'sd6, 16'sd8, 16'sd10, -16'sd28, -16'sd14, 16'sd111,  
    16'sd196, 16'sd111, -16'sd14, -16'sd28, 16'sd10, 16'sd8, -16'sd6, 16'sd0,  
    16'sd2, -16'sd2, 16'sd0, 16'sd2, -16'sd1, 16'sd0, 16'sd1, -16'sd1,  
    16'sd0  
};
```

수정

- logic이 아닌 parameter로 변경하여 해결

```
Information: The register 'shift_reg_reg[27][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[27][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[26][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[25][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[25][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[24][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[24][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[23][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[23][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[22][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[22][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[21][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[21][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[20][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[20][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[19][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[19][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[18][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[18][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[17][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[17][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[16][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[16][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[15][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[15][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[14][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[14][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[13][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[13][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[12][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[12][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[11][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[11][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[10][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[10][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[9][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[9][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[8][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[8][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[7][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[7][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[6][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[6][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[5][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[5][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[4][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[4][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[3][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[3][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[2][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[2][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[1][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[1][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[0][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[0][6]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[32][5]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[32][5]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[31][5]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[31][5]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[30][5]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[30][5]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[29][5]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[29][5]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[28][5]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[28][5]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[27][5]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[27][5]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[26][5]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[26][5]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[25][5]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[25][5]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[24][5]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[24][5]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[23][5]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[23][5]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[22][5]' will be removed. (OPT-1207)  
Information: The register 'shift_reg_reg[22][5]' will be removed. (OPT-1207)
```

PART 8. 고찰



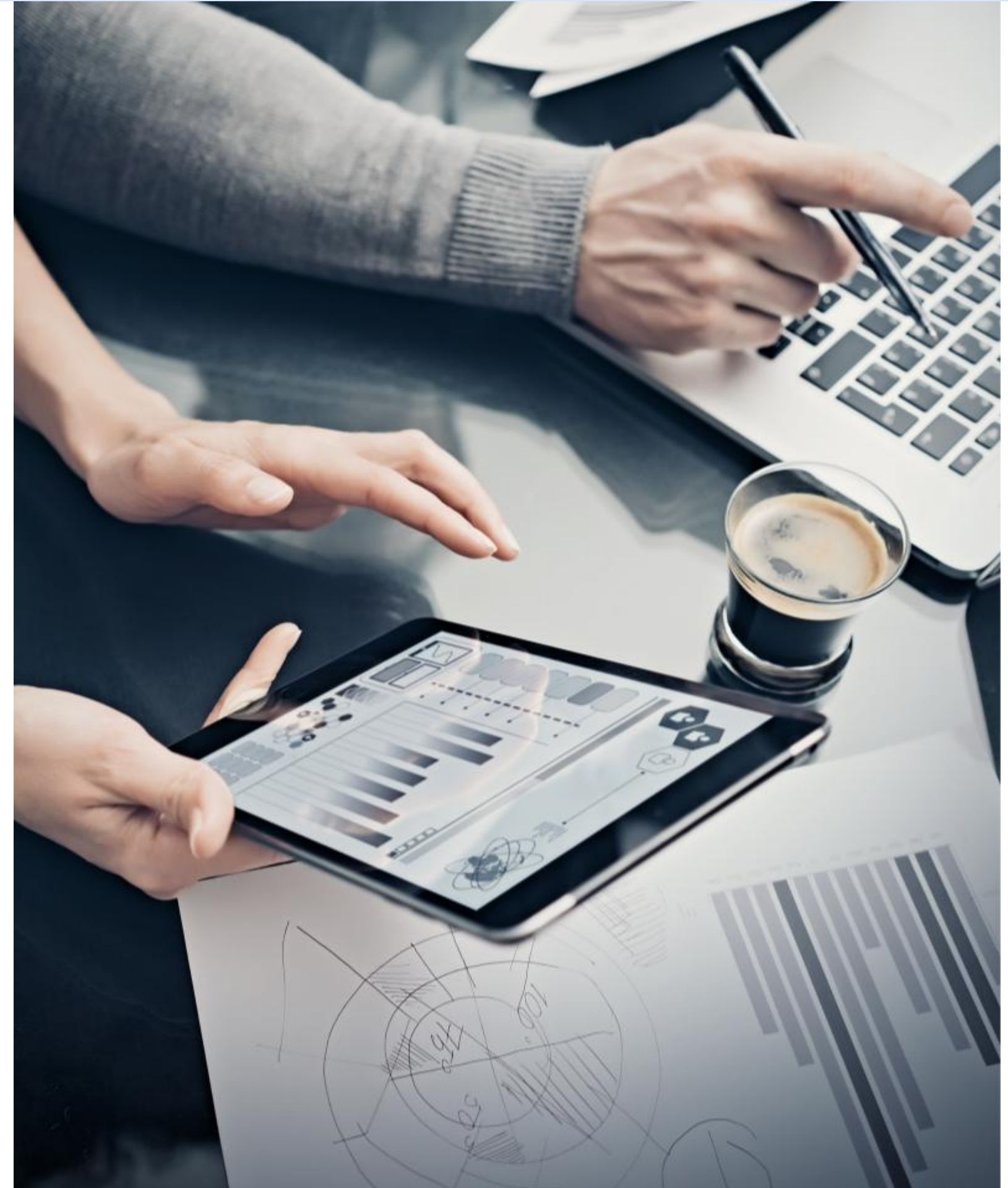
긴글의 텍스트 페이지

해당 페이지는 많은 분량 텍스트를 입력할 때 쓰기 좋은 페이지입니다. 7줄 이상의 텍스트를 넣어야 할 때 이 페이지를 활용해보세요. 중요한 내용이나 길게 의견을 보여주어야 할 경우 해당 페이지를 활용해보세요. 가독성의 핵심은 얼마나 내용을 간결하게 전달하는가입니다. 쉽게 읽을 수 있고, 눈에 잘 띄는 것이 중요해요. 프레젠테이션의 전체적인 톤앤매너와 맞는 폰트와 색상 등을 활용해주세요. 줄 글 중 강조하고 싶은 부분은 볼드체 또는 큰 사이즈로 표현해보세요. 구체적인 예시나 수치 등을 활용하면 좋습니다.

핵심 키워드 A

핵심 키워드 B

핵심 키워드 C



4단 키워드 레이아웃

키워드에 대해 입력해주세요.

- 주제에 대한 세부내용을 간략하게 입력해주세요.



키워드에 대해 입력해주세요.

- 폰트는 프리텐다드 레귤러, 크기는 18pt 입니다.



키워드에 대해 입력해주세요.

- 주제에 대한 세부내용을 간략하게 입력해주세요.



키워드에 대해 입력해주세요.

- 폰트는 프리텐다드 레귤러, 크기는 18pt 입니다.



Goal



목표 키워드를
입력해주세요.

사진이 있는 3단 레이아웃



키워드A에 대해 입력해주세요.

주제에 대한 세부내용을 간략하게 입력해주세요.
폰트는 프리텐다드 레귤러, 크기는 18pt 입니다.
이곳에 위의 사진과 관련된 내용을 입력해주세요.



키워드B에 대해 입력해주세요.

주제에 대한 세부내용을 간략하게 입력해주세요.
폰트는 프리텐다드 레귤러, 크기는 18pt 입니다.
이곳에 위의 사진과 관련된 내용을 입력해주세요.



키워드C에 대해 입력해주세요.

주제에 대한 세부내용을 간략하게 입력해주세요.
폰트는 프리텐다드 레귤러, 크기는 18pt 입니다.
이곳에 위의 사진과 관련된 내용을 입력해주세요.

수치로 설명하는 레이아웃

주요 키워드에 대한 수치를 함께 나타내야할때 사용하면 좋은 페이지입니다. 가독성의 핵심은 얼마나 내용을 간결하게 전달하는가 입니다. 쉽게 읽을 수 있고, 눈에 잘 띄는 것이 중요해요.

45% 키워드 A



30% 키워드 B



25% 키워드 C



키워드 A에 대해 입력해주세요.

주제에 대한 세부내용을 간략하게 입력해주세요.
폰트는 프리텐다드 레귤러, 크기는 18pt 입니다.

키워드B에 대해 입력해주세요.

이곳에 숫자에 대한 내용을 설명해주세요.
주제에 대한 내용을 간략하게 입력해주세요.

키워드C에 대해 입력해주세요.

주제에 대한 세부내용을 간략하게
입력해주세요. 텍스트를 입력해주세요.

2가지 비교 레이아웃

To Be

키워드1에 대해 입력해주세요.

키워드2에 대해 입력해주세요.

키워드3에 대해 입력해주세요.

Keyword 1

Keyword 2

Keyword 3

As - Is

키워드1에 대해 입력해주세요.

키워드2에 대해 입력해주세요.

키워드3에 대해 입력해주세요.

Insight

현재 상황과 이상적인 지향점에 대한 내용을 간략하게 기재해주세요. 폰트는 프리텐다드입니다.

프로세스 레이아웃

Step 1

프로세스 키워드 A

주제에 대한 세부내용을 간략하게
입력해주세요. 폰트는 프리텐다드
레귤러이고 크기는 18pt 입니다.



Step 2

프로세스 키워드 B

주제에 대한 세부내용을 간략하게
입력해주세요. 폰트는 프리텐다드
레귤러이고 크기는 18pt 입니다.



Step 3

프로세스 키워드 C

주제에 대한 세부내용을 간략하게
입력해주세요. 폰트는 프리텐다드
레귤러이고 크기는 18pt 입니다.



Step 4

프로세스 키워드 D

주제에 대한 세부내용을 간략하게
입력해주세요. 폰트는 프리텐다드
레귤러이고 크기는 18pt 입니다.

